



Exploring Human-Centric Dimensions in Blockchain Smart Contracts

HANOUF AL. GHANMI*, Department of Information Systems, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia

SABREEN AHMADJEE, Department of Cybersecurity, College of Computing, Umm Al-Qura University, Mecca, Saudi Arabia

RAMI BAHSOON, Computer Science, University of Birmingham, Birmingham, United Kingdom of Great Britain and Northern Ireland

HAYATULLAHI BOLAJI ADEYEMO, School of Computer Science, University of Birmingham, Birmingham, United Kingdom of Great Britain and Northern Ireland and Faculty of Science, Usmanu Danfodiyo University, Sokoto, Nigeria

Blockchain smart contract technology has revolutionised various industries by automating agreements through immutable and self-executing logic, reducing reliance on third-party intermediaries. However, despite its transformative potential, existing research has predominantly focused on technical aspects—particularly security—while largely neglecting human-in-the-loop concerns. Systematic efforts to explore these concerns from a human perspective have been limited which creates a gap in the literature. This study aims to address this gap by offering a comprehensive understanding of smart contracts from a human-centred perspective. To achieve this, we conducted a systematic literature review to examine human-related issues in smart contracts and their existing solutions. We found that concerns are primarily concentrated in two stages: development and interaction. During the development stage, issues arise in relation to programming languages, including complexity, readability and expressiveness, as well as the legality of smart contracts and their ethical and social implications. In the interaction stage, concerns focus on usability, human readability, trust, governance and cost. Additionally, we identified several quality attributes frequently associated with these concerns such as transparency, accountability, understandability, simplicity, learnability, compliance and fairness. We also uncovered new human-centred quality attributes that are overlooked in existing literature, such as explainability and interpretability. This research offers valuable insights for researchers, requirements engineers and designers by examining existing efforts to address human-centric concerns and proposing future directions and opportunities to improve smart contract design.

CCS Concepts: • **Human-centered computing**;

Additional Key Words and Phrases: Blockchain, Smart Contracts, Human-Centered, Human Concerns

Authors' Contact Information: Hanouf Al. Ghanmi, Department of Information Systems, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Makkah Province, Saudi Arabia; e-mail: halghanmi@hotmail.com; Sabreen Ahmadjee, Department of Cybersecurity, College of Computing, Umm Al-Qura University, Mecca, Makkah Province, Saudi Arabia; e-mail: smahmadjee@uqu.edu.sa; Rami Bahsoon, Computer Science, University of Birmingham, Birmingham, United Kingdom of Great Britain and Northern Ireland; e-mail: r.bahsoon@cs.bham.ac.uk; Hayatullahi Bolaji Adeyemo, School of Computer Science, University of Birmingham, Birmingham, United Kingdom of Great Britain and Northern Ireland and Faculty of Science, Usmanu Danfodiyo University, Sokoto, Sokoto, Nigeria; e-mail: adeyemo.hayatullahi@udusok.edu.ng.

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1 Introduction

Blockchain technology has revolutionised how agreements are executed by introducing smart contracts as a cornerstone of this innovation. These smart contracts, essentially self-executing agreements encoded on blockchain platforms such as Ethereum [27], leverage the blockchain's distributed and immutable nature to eliminate the need for third-party intermediaries [3, 67, 79]. Smart contracts aim to enhance trust, transparency and efficiency. However, their unique features—immutability, automation and enforceability—present significant challenges that differ from traditional software development. These characteristics demand an extremely high level of precision and security in the code as any errors become permanent upon deployment and can result in substantial financial losses [72, 100, 101]. As a result, the research landscape around smart contracts is predominantly focused on addressing these technical and security concerns [1, 120, 132].

Despite this technical focus, there is a growing recognition of the importance of human-centric approaches in smart contracts. Stakeholders encounter numerous challenges, from development hurdles to complex user interactions, which hinder the broader adoption of this technology [52]. Integrating smart contracts into real-world applications requires a deep understanding of human in the loop issues. Effectively integrating smart contracts into real-world applications requires a nuanced understanding of human-in-the-loop issues. In response, this study conducts a systematic literature review (SLR) to examine how human-centric concerns are addressed in the context of smart contracts. In this context, “human-centric” refers to concerns expressed from the perspectives of stakeholders—such as users, developers, and decision-makers—including their needs, expectations, and interactions with smart contract systems. Rather than viewing smart contracts solely as technical artefacts, a human-centric approach recognises the diverse roles and expectations of stakeholders. This perspective includes key concerns such as usability, understandability and trust, which are critical for all stakeholders, including non-technical users and developers.

Specifically, we seek to explore the question: *“What is the current state of the art in addressing human-centred concerns in blockchain smart contracts, and what are the associated quality attributes?”* Through the SLR, we examined 77 primary studies, identifying key stakeholder concerns and the corresponding solutions proposed in the literature. These concerns are categorised into two main stages from the stakeholders' perspective: *Development* and *interaction*. Development concerns are grouped into three categories: language, legality, and ethical/social implications. Language emerged as the most frequently reported concern with issues related to complexity, code-readability and expressiveness, affecting both technical and non-technical stakeholders involved in the development. Interaction concerns include usability, human-readability, governance, trust and cost, all of which impact end-users and are influenced by contract design decisions.

Additionally, we explore the current state of solutions and interventions addressing these concerns. Our collection includes two main types of interventions: *analytical studies* and *human-centric approaches*. The literature investigates human concerns through methods such as human-based surveys, empirical studies, conceptual and exploratory studies and comparative analyses. Given the infancy of this technology, this group provides new insights into the current understanding of human concerns. The proposed human-centric approaches are scattered across different domains such as new languages and third-party tools. Therefore, we categorise these solutions into new *languages and modelling approaches*, *code-comment methods*, *visualisation tools*, *natural language solutions* and *detection and assessment tools*. This classification aims to clarify existing efforts to aid researchers in identifying available resources and solutions.

Furthermore, we identify quality attributes derived from the human concerns by mapping them to the ISO systems and software engineering standard [57], highlighting the need for greater attention in areas such as understandability, transparency, accountability, simplicity, learnability, usability and accessibility. We also examine the trustworthiness characteristics of the National Institute of Standards and Technology (NIST) [61], highlighting explainability and interpretability as overlooked quality attributes.

As a result, we present several research directions to assist researchers in advancing human-centric smart contracts. Our main aim is to understand the current state, identify gaps and provide insights that can drive further innovation in this domain. Accordingly, the main contributions of this study are as follows:

- We present a classification of concerns from stakeholders' perspectives, structured into two main stages: development and interaction. During development, key concerns include language (complexity, code readability, and expressiveness), legality, and ethical and social implications. In the interaction stage, the focus shifts to usability, human readability, trust, governance and costs.
- We provide a categorisation of scattered solutions, comprising human-centred approaches and analytical studies aimed at integrating human considerations into the development and interaction phases of smart contracts.
- We identify commonly reported human-centred quality attributes and explore additional quality attributes that may have been overlooked in the existing literature.
- We highlight unexplored gaps and opportunities in human-centric smart contract design that require further investigation.

The remainder of this article is organised as follows: Section 2 presents a brief overview of blockchain smart contracts and their applications. Section 3 provides insight into the research methodology used for conducting the systematic literature review. Sections 4 and Section 5 offer the results and findings of the review, addressing the research questions. Section 6 discusses identified gaps, future directions and potential threats to validity. Finally, Section 7 compares our study with related work, followed by a conclusion in Section 8.

2 Background

This section provides an overview of the fundamental concepts of blockchain and smart contracts, and explores their applications.

2.1 Overview of Blockchains

Blockchain technology is a decentralised ledger system that records transactions across a network of computers in a secure and transparent manner [67, 106]. The term “blockchain” is describing exactly the definition which is referred to as a series of digital blocks that are connected to each other by a reference hash. Each block has the reference of the previous block creating a chain of blocks [134]. A transaction is created to record the assets exchanged or record the outcome of executing a function of a smart contract. If transactions have been accepted and added into a block, they cannot be updated or modified. The reason for transactions being immutable is to ensure the integrity of transactions [3]. The literature highlights three key characteristics of blockchains [49, 50, 135]: immutability, which ensures data cannot be changed or deleted; decentralisation, which operates on a distributed network making it resistant to tampering; and transparency, allowing all participants to view and verify transactions.

Blockchains gained prominence with the emergence of Bitcoin in 2009 [28]. Various blockchain platforms have been developed to support a wide range of use cases beyond cryptocurrency. Among these, Ethereum [27], which is the most popular blockchain platform that supports the development and execution of smart contracts. Unlike Bitcoin, which primarily serves as a digital currency, Ethereum's architecture facilitates the deployment of a wide range of decentralised applications (DApps) within its network. This versatility is underscored by Ethereum's support for multiple high-level programming languages, notably Solidity [105], allowing developers to create and deploy applications tailored to different domains [120].

2.2 Overview of Smart Contracts

Smart contracts are computer programs stored on a blockchain or distributed ledger system that automatically execute specific operations when predefined conditions are met [58]. They provide a mechanism to automate and enforce processes and conditions, reducing or eliminating the need for intermediaries [78, 79]. The term "smart contract" has been widely debated. Vitalik Buterin, one of the creators of Ethereum, has remarked that the term might be misleading, suggesting alternatives such as "persistent scripts" to better reflect their technical nature ¹. On the other hand, according to the ISO/TR 23455 Blockchain standards, a smart contract might represent terms in a legal contract, creating enforceable obligations under specific jurisdictions [58]. From this perspective, smart contracts can also be defined as agreements with predefined terms written directly into their code [67, 108]. Thus, smart contracts can be broadly understood as programs or scripts running on blockchain systems. While some use cases focus on representing legal agreements, others extend to automating various tasks without necessarily involving legal terms. Consequently, smart contracts take multiple forms depending on their context—whether legal, technical or operational.

Smart contracts have significantly accelerated blockchain adoption by enabling computational tasks similar to object-oriented programming languages. While digital agreements and technology-driven rule enforcement are not new, smart contracts stand out due to their unique combination of asset control, automation, immutability, and enforceability. A smart contract typically consists of three main elements: storage, balance and program code. The contract's state comprises its storage and balance. Once deployed on the blockchain, the contract becomes immutable and is assigned a unique address. However, this immutability applies unless an upgradability pattern is explicitly coded, allowing the contract to be modified or replaced under predefined conditions [55]. Blockchain users, typically through externally owned accounts (EOAs) or other smart contracts, are responsible for creating and deploying smart contracts. Miners, on the other hand, play a critical role in recording transactions, including those that involve contract creation, into blocks through consensus mechanisms [87]. Based on the transactions received, smart contracts can read from or write to their storage, send or receive messages or funds, and alter their state within the blockchain network.

Smart contracts are widely used in various DApps, as discussed in the literature, with commonly mentioned terms such as DAO (Decentralised Autonomous Organisation) and NFTs (Non-Fungible Tokens). For clarity, these terms will be consistently used throughout this study to refer to specific applications of smart contracts. DAOs are entities managed by smart contracts, where decision-making processes and governance are automated through code [119]. NFTs are distinct digital assets that cannot be divided or duplicated, making them ideal for representing ownership of digital art, collectibles and other unique items. Smart contracts are employed to create, transfer and manage NFTs, ensuring a secure and transparent way to verify ownership and authenticity [96]. In addition, smart contracts in cryptocurrencies facilitate automated trading strategies, manage processes within DeFi (Decentralised Finance) protocols, enable asset transfers and allow for assets tokenisation [102].

3 Research Methodology

This study employed an SLR methodology to thoroughly investigate our specific research area. Following the well-established methodology by Kitchenham and Charters [68], we meticulously proceeded through several key stages: (i) identifying research questions, (ii) devising a search strategy, (iii) establishing inclusion and exclusion criteria, (iv) conducting a rigorous study selection process, (v) assessing the quality of selected studies, and (vi) extracting and analysing relevant data. This methodology enabled us to comprehensively assess the current state of the art and identify potential research gaps. Figure 1 illustrates our SLR research process.

¹<https://x.com/VitalikButerin/status/1051160932699770882>

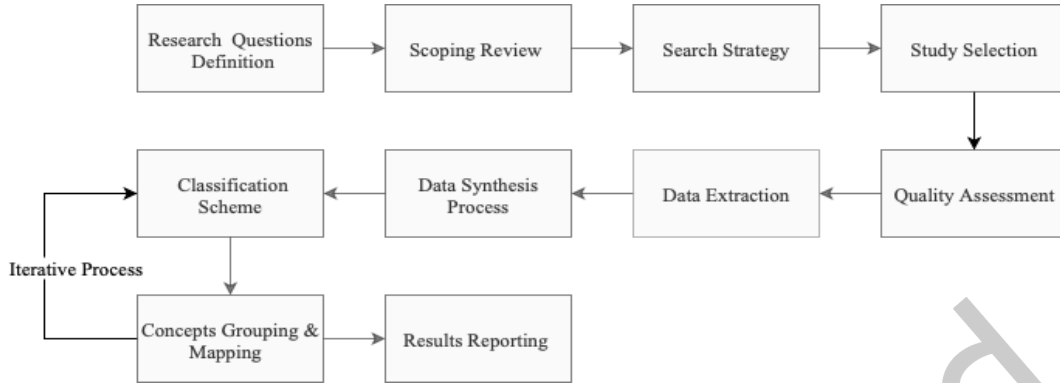


Fig. 1. Systematic Literature Review Process

3.1 Research Questions

To conduct the SLR, we divided the primary research question— “*What is the current state of the art in addressing human-centered concerns in blockchain smart contracts, and what are the associated quality attributes?*”—into more specific sub-questions to guide this investigation:

- **RQ1:** *What are the most commonly reported concerns regarding blockchain smart contracts from a human perspective? and how are these concerns currently being addressed?*
- **RQ2:** *How can we identify quality attributes commonly associated with these human-centred concerns?*

3.2 Search Strategy

To gather relevant studies, we conducted a comprehensive literature search across several well-established academic databases and digital libraries including IEEEExplore, ACM Digital Library, Web of Science and Scopus. These databases cover publications from publishers such as Elsevier and Springer, ensuring a diverse and extensive collection of relevant literature. The search string keywords were developed based on insights from a scoping review, which involved evaluating potential keywords. During this review, we realised that focusing solely on terms related to human-centred was too restrictive. For example, when we searched for ("human-centric" AND "smart contracts") in IEEE Explore, we found fewer than ten results. To address this limitation, we expanded our search criteria to include additional relevant terms, particularly those related to legal, ethical and social concerns.

Furthermore, the literature contains various terms or combinations related to quality attributes (e.g., quality concern, non-functional requirements, concern, QAs or NFR). Many studies also discuss specific quality attributes such as privacy, performance or transparency without explicitly referencing the term "quality". We experimented with different search strategies related to quality attributes but found that they retrieved studies unrelated to our research questions. Therefore, to avoid missing any relevant studies, we used the search string in a generic form and removed any terms related to concerns and qualities similar to [71]. As a result, the tailored search strategy specifically focused on blockchain smart contracts with consideration for humans in the loop as follows:

- *(Blockchain) AND ("smart contracts" or "smart contract") AND (human OR human-centred OR human-centric OR socio-technical OR ethical OR ethics OR legal OR compliance OR regulation OR social) AND (design OR requirement OR patterns OR specification)*

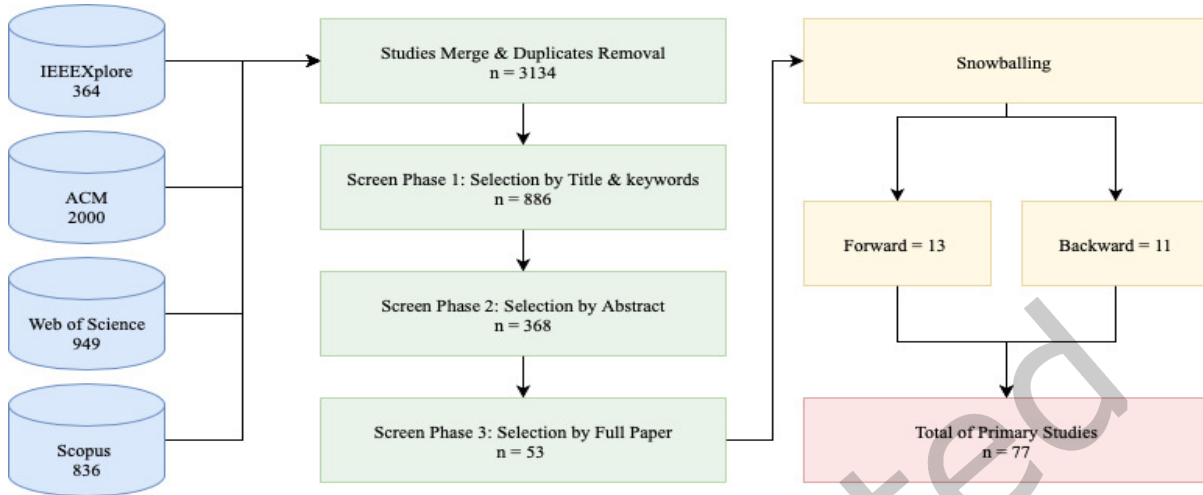


Fig. 2. An Overview of the Selection Process

3.3 Study Selection

To ensure the relevance of the papers retrieved from the search, a screening process was developed. This process involved the identification and application of inclusion and exclusion criteria to ensure objectivity in the outcomes, as follows:

3.3.1 Inclusion Criteria (I).

- **I1:** Papers are peer-reviewed journals, conferences, book chapters or workshops.
- **I2:** Papers include information about blockchain smart contracts, focusing on concerns arising from human perspectives and considerations.
- **I3:** Papers explicitly relate to the topics of research questions, focusing on human concerns and proposing or discussing solutions, methods, frameworks or approaches for addressing the identified concerns and qualities.

3.3.2 Exclusion Criteria (E).

- **E1:** Papers discuss blockchain with technologies such as IoT, cloud environments, mobile platforms and disciplines other than computer science.
- **E2:** Papers propose the utilisation of blockchain and smart contracts as solutions for specific use cases and applications.
- **E3:** Papers address concerns related to blockchain properties and have limited discussions on smart contracts.
- **E4:** Papers with restricted access or unavailable full text.
- **E5:** Papers written in languages other than English.

3.3.3 Screening Phases. The final selection of research papers was determined through multiple rounds of filtering. Figure 2 illustrates the search and selection processes.

Initial Search: A total of 4194 relevant studies were retrieved from IEEE Explore, ACM Digital Library, Web of Science and Scopus using the search string defined in Section 3.2. The initial search on the ACM Digital Library

Table 1. Papers Quality Assessment Criteria [130]

Characteristic	QA Criteria	ID
Rationality	Are the research aims/objectives clearly defined?	Q1
	Is the study context clearly outlined?	Q2
	Is the paper founded on research?	Q3
Rigor	Does the method adequately address the research aims?	Q4
	Is the data collection method adequately described?	Q5
	Is the data analysis sufficiently rigorous?	Q6
Credibility	Is there a clear description of findings?	Q7
	Do the researchers discuss any limitations and threats to the validity of their findings?	Q8

yielded over 2,000 results. However, due to limitations in the library's retrieval process, only the first 2,000 studies were successfully retrieved.

Duplicates removal: After eliminating duplicate entries, the initial pool of studies was refined to 3134 unique results.

First Round: Studies were evaluated based on their titles and keywords to exclude irrelevant studies. This screening phase involved the application of criteria I1, E1 and E2, which led to the selection of 886 studies meeting our initial selection criteria. However, certain studies posed challenges in determining their relevance based solely on titles and keywords, prompting us to advance them to the following screening phase for further assessment.

Second Round: The authors independently reviewed the abstracts of every paper that passed from the previous phase. We applied criteria I2 and E3 to screen each paper resulting in 368 studies. Papers lacking clarity in their abstracts regarding a comprehensive discussion on smart contracts were advanced to the following screening phase.

Third Round: The authors reviewed the full text of the papers selected in the previous round to finalise the list. Criteria I3, E4 and E5 were applied during this stage, resulting in the final selection of 53 primary studies for in-depth analysis to address the research questions.

Snowballing: We implemented a snowballing strategy following Wohlin's approach [123], which involved both backward and forward snowballing. This strategy expanded our search by scrutinising the references of the initially selected papers (backward) and their citations (forward). Through multiple iterations of both forward and backward snowballing, and by applying inclusion and exclusion criteria, we identified additional studies that met our criteria. This process resulted in the inclusion of 24 more studies, bringing the total to 77 studies.

During our screening process, we used Cohen's Kappa (k) statistic to measure the level of agreement between authors [107], achieving a k value of 0.762, which indicates substantial agreement according to established benchmarks [107]. In cases of disagreement, authors discussed viewpoints to resolve conflicts. This process ensured that the selected studies met our selection criteria and maintained the integrity and objectivity of our final selection of the 77 studies.

Table 2. Data Extraction Form

ID	Item	Focus/Insight
D1	Study ID	Identification
D2	Study Title	Demographic
D3	Publication Year	Demographic
D4	Publication Type	Demographic
D5	Publisher	Demographic
D6	Concern Expressed	RQ1
D7	Proposed Solution or Discussion	RQ1
D8	Quality Attributes	RQ2

3.4 Quality Assessment

To evaluate the quality and strength of evidence for each included primary study, we conducted a quality assessment based on the framework established by Yang et al [130]. We adapted the commonly used criteria for quality assessment in software engineering to suit our needs. These criteria covered aspects of rationality, rigour and credibility, resulting in the selection of eight specific evaluation questions, as presented in Table 1.

Authors evaluated each primary study by answering questions using a matrix of values: 0 for "no," 0.5 for "to some extent," and 1 for "yes." The final quality score for each study was determined by summing these values. In cases of discrepancies between authors, discussions were initiated to reassess the studies and establish the final score. These scores evaluate the quality of primary studies and are not used as grounds for excluding any study from consideration.

Our SLR inclusion criteria specified that only peer-reviewed studies were considered for this review. Consequently, all the evidence collected met a baseline standard of academic rigour. Although quality scores were not used as exclusion criteria, they informed the synthesis process by guiding the classification of findings. Studies with higher scores were given greater consideration during the development of thematic categories, whereas lower-scoring studies were used to support and enrich those classifications. More details on the scoring criteria are provided in the Results section 4.2.

3.5 Data Extraction

Data extraction involves gathering the necessary data items to analyse the studies. Table 2 was specifically designed to rigorously extract the information required to address the research questions of this study. Data items D1 to D5 contain demographic information intended for quantitative analysis to demonstrate basic information about the studies. Conversely, items D6 to D8 are relevant to the research questions, requiring in-depth investigation through qualitative analysis. We utilised NVivo [82], a software tool widely used for its text and word analysis capabilities, to identify the raw data needed to answer the research questions. NVivo facilitates the coding process and provides easy reference for extracting information, which can be shared among authors.

3.6 Data Synthesis

Our analysis of the primary studies data focuses on the qualitative information collected from the data extraction phase. To address RQ1, we started by defining the term 'concern' according to [57], where it is described as an aspect of a problem or consideration that is important or affects one or more stakeholders. We utilised thematic analysis [17] to navigate the data effectively within its context. The thematic analysis involves six distinct phases: (i) familiarising with the data, (ii) generating initial codes, (iii) themes searching, (iv) reviewing themes, (v) defining

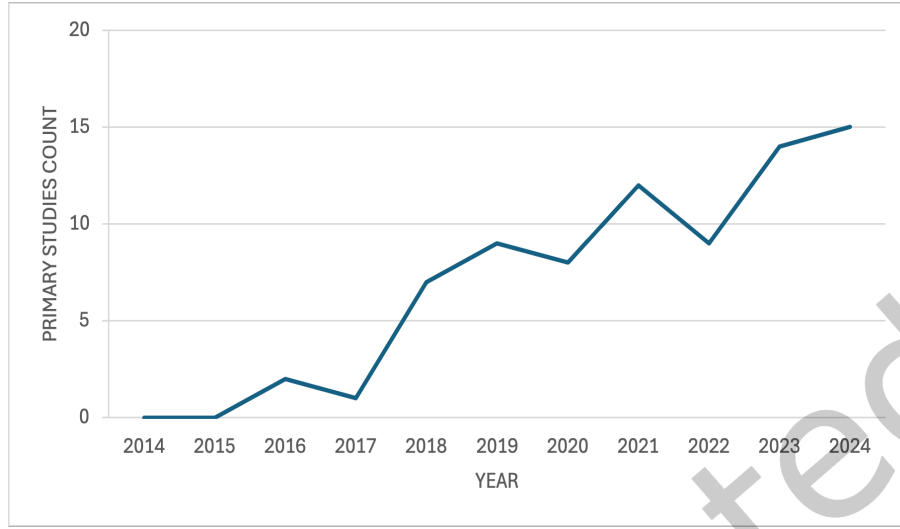


Fig. 3. Increasing trend in smart contract publications over the years

and naming themes and (vi) reporting. After adopting the coding techniques outlined in [98], we searched for themes once all data have been coded. During this phase, we analysed the relationships between codes and compiled relevant data within themes to identify key segments for addressing the first research question.

To address RQ2, we utilised two data synthesis processes. Firstly, we adopted the ISO/IEC/IEEE 24765 [57] standards as terminology guidelines to identify common qualities associated with concerns in smart contract development and interaction. This standard was chosen for its comprehensive collection of standardised terminologies contributed by the International Organisation for Standardisation (ISO), the International Electrotechnical Commission (IEC) and IEEE Standards serving as a valuable reference for systems and software engineering standards. The compiled list of definitions established a foundation for identifying the qualities found in the literature. Additionally, we reviewed ISO/TR 23455:2019 Blockchain and distributed ledger technologies [58] but noted that it does not provide definitions for system quality attributes (e.g., transparency, visibility and accountability) in their context.

Studies may not explicitly mention quality attributes. For instance, studies referring to safety may use terms such as "safe" or "safer," and accessibility may be referred to as "accessible." We used NVivo tool [82] to code segments and phrases related to these quality attributes. To be considered, a term must appear at least twice ($n \geq 2$) in the study, ensuring consistent recognition as a quality attribute. This approach prevents inflating the importance of less frequently mentioned terms, focusing on the commonly reported qualities. Subsequently, we mapped quality definitions with extracted terms.

The second synthesis aims to uncover quality attributes that may have been overlooked or received limited attention in the smart contracts literature. Recently, the concept of "trustworthiness" has gained attention in emerging technologies such as artificial intelligence (AI) and machine learning (ML). Trustworthy systems must meet stakeholders' expectations by demonstrating characteristics such as accountability, authenticity, availability, integrity, safety, transparency and usability [60], which align with our objectives. Therefore, adhering to the characteristics of trustworthy systems can systematically help identify overlooked quality attributes in smart contracts. While several guidelines and standards explore these characteristics [59, 60], we selected the trustworthiness characteristics from the NIST [61]. This choice was influenced by the widespread adoption of

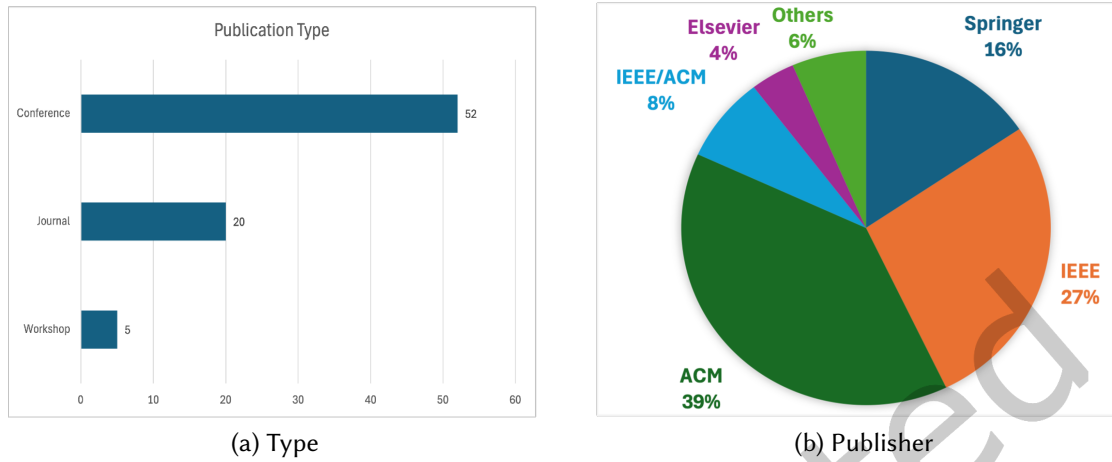


Fig. 4. Distribution of Types and Publishers Among Selected Papers

this framework in various systems literature and its recent publication. The framework provides a structured approach outlining seven main characteristics of trustworthiness, each with its quality attributes, allowing for clear identification.

4 Results

This section provides an overview of the descriptive metadata of primary studies, including demographics and quality scores.

4.1 Demography of Studies

The demographic details of the studies provide essential information about the selected papers, including publication year, study type (e.g., journal, conference or workshop) and publishers. In this section, we present these details for the 77 selected studies to statistically describe their characteristics and support the quantitative analysis of the review.

4.1.1 Publication year. Figure 3 presents the distribution of publication years for the selected primary studies. Interest in smart contracts began to grow noticeably after 2016, following the launch of Ethereum in late 2015 [78]. Since then, the number of publications has shown a steady upward trend. In particular, the proportion of studies increased from 12% in 2021 to 22% in 2024. This growth reflects the rising academic and industry attention toward smart contracts and their expanding applications, particularly in decentralised domains such as DeFi, NFTs, and DAOs. The literature search was last updated on December 31, 2024, and Figure 3 captures publications up to that date.

4.1.2 Type and Publisher. Our study collection exclusively includes peer-reviewed articles. Figure 4 offers insights into the distribution of publication types among the selected papers, highlighting conferences (52 studies), journals (20 studies) and workshops (5 studies). Furthermore, Figure ?? showcases the publishers of the selected studies. The majority of papers are published by ACM (39%), followed by IEEE (27%) and Springer (16%). Other publishers include IEEE/ACM collaborations (8%), Elsevier (4%), and the remaining 8% from various publishers such as Taylor & Francis, IBM, and Sage Journals. The data presented in Figure 4 represent the distribution of publication

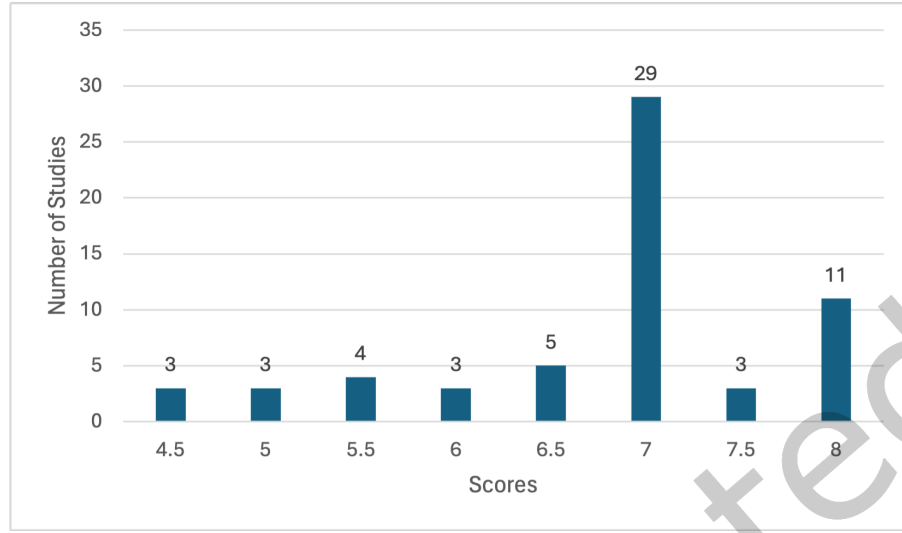


Fig. 5. Distribution of Quality Score Among Selected Studies

types and publishers for the 77 primary studies included in this review, covering publications up to December 2024.

4.2 Quality Scores Results

The quality scores assigned to the primary studies were derived using the assessment criteria detailed in Section 3.4. Each study received a total score ranging from 4.5 to 8, with intervals of 0.5. Only three studies received the lowest score of 4.5, while most studies scored between 7 and 8, as shown in Figure 5.

Among the eight quality assessment criteria highlighted in Table 1, Q8 received the lowest average score, indicating that many studies provided limited or no discussion of their limitations and threats to validity. Similarly, Q5, which evaluates the adequacy of data collection methods, negatively impacted the scores of several studies due to a lack of detail regarding how data were collected.

These scores informed the synthesis process by indicating the relative strength of evidence presented by individual studies. In particular, findings from studies with higher scores (7–8) were given greater weight during theme development and when addressing conflicting evidence. For example, studies such as [37], [15], and [127]—all scoring in the higher range—were instrumental in identifying "trust" and "governance" as distinct thematic categories. Nonetheless, the study by [70], which received a lower score of 4.5, contributed supplementary evidence relevant to both categories, thereby enriching the synthesis despite its lower quality rating. Appendix A presents the complete results of the quality assessment for all included studies.

5 Analysis of the Selected Studies

In this section, we provide a detailed analysis to answer the research questions. Section 5.1 and 5.2 discuss the findings related to answering RQ1, which explores the commonly reported concerns and the existing solutions. In Section 5.3, we address RQ2 by identifying both current and overlooked qualities for designing human-centric smart contracts. Figure 6 presents the layout of this section along with a summary of the findings and classifications. Detailed information on each study's concerns and solutions is provided in the Appendix B.

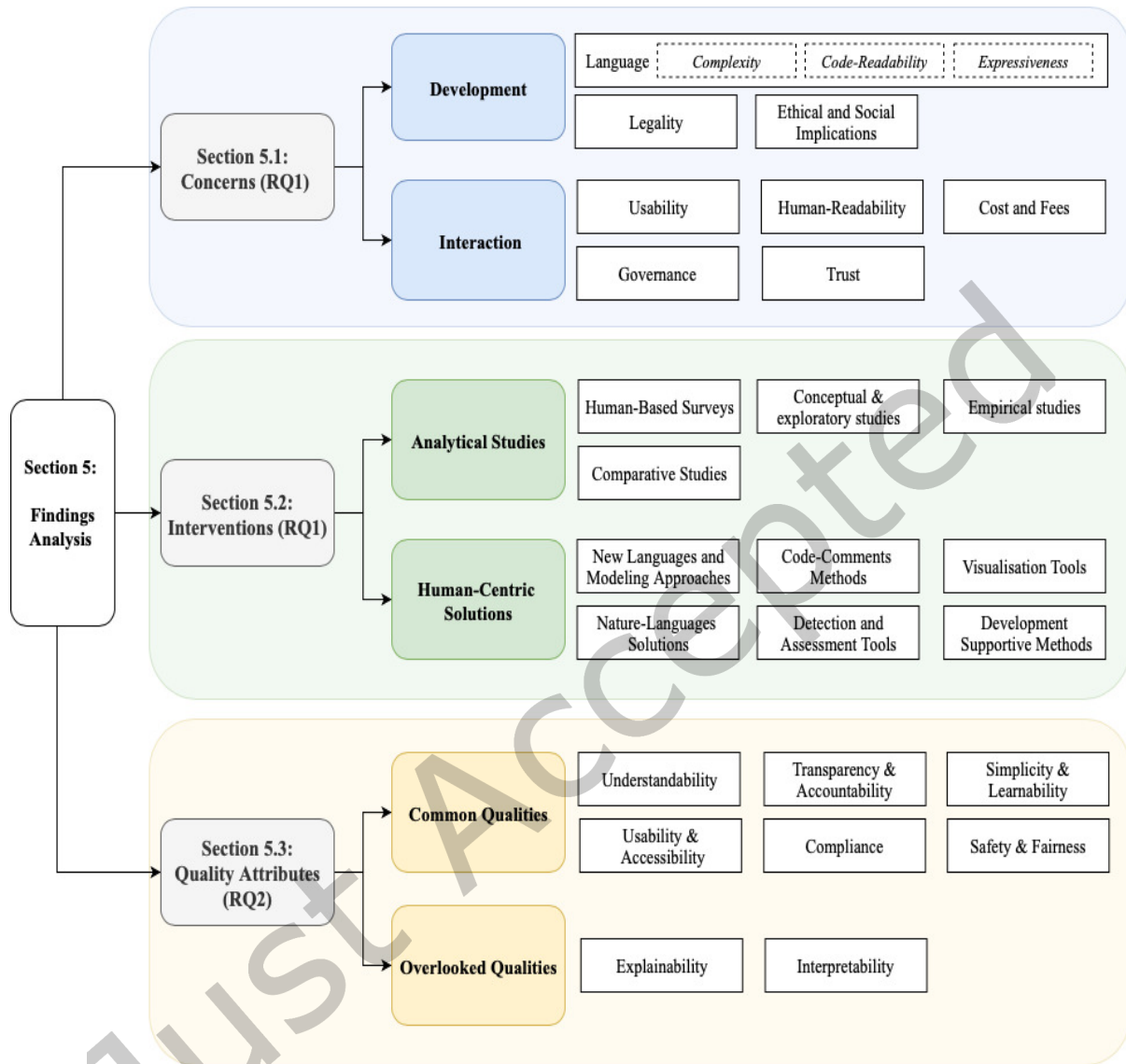


Fig. 6. An Overview of Section 5 Structure Presenting Findings Analysis and Classifications to Answer the Research Questions

5.1 Common Human Concerns in Smart Contract (RQ1)

The main concerns identified in the primary studies based on the iterative process employed [17] revolve around two key stages: (i) Development and (ii) Interaction, which are viewed from the perspectives of different stakeholders. The development stage impacts various stakeholders, including IT background developers, non-IT background developers and domain experts. Interacting with smart contracts mainly focuses on end-users. In the development category, concerns are classified into language, legality and social and ethical concerns. In the

interaction category, concerns are classified into usability, human readability, governance, trust and cost. Table 3 presents the main concerns along with their classification and corresponding studies.

5.1.1 Development Human Concerns.

Language: The most concern reported within our primary collection revolves around programming language-related considerations and the coding process. Given that smart contracts can embody contractual agreements, using code to represent these agreements poses challenges in interpretation. This complexity is further compounded by using imperative programming languages such as Solidity [105], which inherently limit the expression of legal clauses. Therefore, comprehending the code in terms of its embedded terms and policies becomes a significant hurdle. As a result, these language concerns critically impact developers and domain experts during the creation and development processes. We categorise them into key subcategories: Complexity, code-readability and expressiveness. Two secondary studies have performed SLRs to analyse the current landscape of smart contract languages, focusing on identifying the various languages and their characteristics [24, 117].

- **Complexity** refers to the intricacy and difficulty associated with the programming language and the resulting code. It considers factors such as syntax intricacies, limited features, lack of standardised development practices and the learning curve required to master the language [42, 99]. A complex language can impede the achievement of required goals and policies set by the system's stakeholders [114], leading to convoluted code structures and difficulties in comprehension and maintenance [46]. With limited features, developers often adapt their requirements to the rigid structures, resulting in error-prone code and higher cost [13, 74, 111, 125]. Additionally, many development tools are still immature and, combined with limited developer knowledge or awareness, fail to adequately address developers' needs [10, 54, 121]. As a result, developers are faced with the daunting task of manually implementing and enforcing terms within the constraints of the existing languages, tools and libraries misuse, leading to increased complexity and potential risks [54, 84, 86]. In the primary studies, three perspectives are discussed regarding concerns related to language complexity: IT background developers, non-IT background developers, and collaborative developments.
 - *IT-background developers:* The complexity of generating or creating code can greatly affect how easily developers can work with and deploy smart contracts. The main concern revolves around the usability and complexity of programming languages and tools, which often demand significant time and expertise to navigate effectively [12, 13, 16, 42–44, 46, 90, 99, 111]. Balancing security and usability of the languages is particularly challenging when writing code in languages such as Solidity [86]. In addition, complexity is further compounded when developers design smart contracts for different blockchain platforms [43, 44].
 - *Non-IT background developers:* The main concern is the complexity involved in smart contract development, which primarily caters to developers. These systems lack development usability for developers without programming experience, making it challenging for them to create smart contracts. Legal professionals and developers unfamiliar with smart contract technology face difficulties understanding and working with these contracts [16, 33, 42, 76, 93, 109, 122, 125]. There is a recognised need for natural language to simplify the creation process, enabling non-programmers to participate in smart contract development.
 - *Collaborative Development:* The complexity of collaborative development with varying expertise and perspectives can lead to communication breakdowns and conflicts in drafting contract terms and conditions [48, 89, 112]. Semantic consistency poses a challenge, as contracting parties may interpret terms differently based on their contexts. Achieving common consent is difficult due to variations in natural language grammar, particularly in translations between languages, which may alter the original meanings of contract clauses. Furthermore, the need for multilingual understanding arises in the globalised business landscape where companies operate in multiple markets and require contract interpretations across

Table 3. A Summary of Human Concerns in Smart Contracts

Stage	Category	Concern	Corresponding Studies
Development	Complexity	Concerns regarding the intricacy of a smart contract's code, especially when its structure or logic creates challenges in comprehension.	[12, 13, 16, 33, 42–44, 46, 48, 76, 86, 89, 90, 99, 109, 111, 112, 114, 122, 125]
	Code-Readability	The lack of clarity and ease of understanding makes it difficult for developers to interpret, maintain, debug and reuse the code effectively.	[8, 45, 116, 129, 133]
	Language Expressiveness	Challenges in ensuring that a smart contract's code can clearly and accurately convey its intended functionality, including the ability to express complex business logic and translate it into executable code.	[2, 11, 16, 23, 32, 41, 69, 92, 94, 103, 111–113, 125]
	Legality	Concern over the adherence of a smart contract to relevant legal regulations and contractual obligations.	[19, 22, 66, 69, 77, 81, 113]
	Ethical & Social Concerns	The lack of consideration of broader societal and ethical implications in the design and implementation of smart contracts, including issues related to fairness and accountability.	[26, 73, 118]
Interaction	Usability	Concerns regarding the limited usability of smart contracts, particularly in terms of interaction, navigation and comprehension.	[6, 34, 35, 51, 63–65, 75, 80, 85]
	Human-Readability	Concern over the lack of accessible language and structure in smart contracts, making them difficult to understand for individuals with varying levels of technical expertise.	[33, 89, 94, 133]
	Governance	Concern pertains to the extent of governance and decision control mechanisms, including centralised risks.	[37, 62, 70, 128]
	Cost & Fees	The challenges users face in comprehending and managing transaction expenses, particularly the complexities in understanding fee structures.	[29, 34, 35, 63, 83]
	Trust	Concerns regarding users' confidence and trust in the reliability of a smart contract's operations, code and stakeholders.	[7, 15, 34, 35, 37, 70]

different languages. These challenges underscore the intricate nature of collaboratively developing smart contracts.

- **Code-Readability:** The literature highlights two key concerns of readability: code-readability and human-readability. Code readability refers to how well developers can understand the code for maintenance and reuse, while human readability refers to how easily stakeholders can comprehend the code. In decentralised applications, the smart contract bytecode is visible to all users, whereas the source code is sometimes not available because its publication is not required for the contract to function. However, in ecosystems such as Ethereum, releasing the source code has become common practice as a trust-building measure, enabling developers and stakeholders to read, assess and understand the contract for interaction. Therefore, human-readability reflects an interaction aspect discussed in the next subsection. The challenge of ensuring code-readability in smart contract development is particularly pronounced due to the extensive reuse of code [8, 116]. It has been stated that 10% of the security vulnerabilities are related to code reuse that has no comments [129]. This challenge is further compounded by code optimisation to minimise gas consumption. Enhancing readability may result in higher deployment and execution expenses, whereas a lack of readability can lead to errors and hinder developers to maintain and reuse code [8, 116]. Furthermore, due to code reuse practices, comments often contain inconsistencies that can mislead developers and users, potentially introducing vulnerabilities to contracts [45]. The lack of effective comments in most smart contracts code is concerning in terms of code-readability [129, 133].
- **Expressiveness:** It highlights the language's ability to enable developers to articulate complex concepts or solutions concisely through its features and structures. The expressiveness concerns inherent in existing languages present significant challenges, particularly in accurately converting natural language contracts into machine-readable code while preserving validity and semantic fidelity [11, 16, 32, 41, 94, 111, 125]. This challenge is especially pronounced in the development of legal smart contracts, where a thorough understanding of legal contract terms is essential for their precise translation into software requirements [2, 23, 41, 69, 92–94, 111, 112]. The translation process involves mapping the content and structure of legal contracts to a formal representation understandable and executable by smart contract systems, demanding collaboration between engineers and domain experts to ensure that contract terms are expressed accurately into executable code. However, increasing the expressiveness of the code can compromise the safety of smart contracts [103]. Additionally, the study [113] argued that the focus should not be on the translation and expressiveness of programming languages but rather on the design of smart contracts.

Legality of Smart Contracts: One of the primary concerns highlighted in developing smart contracts is the absence of explicit legal regulations across many jurisdictions. This lack of clarity raises uncertainties regarding their legal validity and enforceability [19, 22, 66, 69, 77, 81]. Smart contracts, functioning as legally binding agreements, often encounter challenges in harmonising with established legal frameworks such as international law, securities law and general data protection regulations (GDPR) [22]. For example, smart contracts, inheriting immutability from blockchain technology, encounter challenges in adhering to the GDPR's "right to be forgotten" principle. This discrepancy leads to enforceability and legality issues, with potential implications for consumer protection and transactional clarity. Moreover, traditional contract law tools, designed for conventional settings, may not seamlessly adapt to the technological complexities and immutability of smart contracts such as termination, rescission, modification and reformation [66, 77]. Ongoing debates on smart contracts vary widely, with some disputing their classification as contracts altogether or perceiving them as a disruptive force in traditional contract law [19, 113].

Ethical and Social Implications: Few studies have shed light on ethical and social concerns during the development of blockchain and smart contracts technology. These concerns include encoded biases, transparency, accountability in governance and decision mechanisms and the risks associated with commodifying social

interactions and values [9, 14, 26, 73, 118]. The lack of public awareness and transparency in decision-making processes within smart contract systems leads to biases and may empower specific parties while excluding others. Although the bytecode of smart contracts is publicly accessible, the opacity in how design choices, update mechanisms and governance decisions are made can obscure the nuanced social interactions that shape contractual relationships. This highlights the increasing need for human-centric, sustainable and resilient smart contracts that better adapt to stakeholder and environmental needs [31].

5.1.2 Interaction Human Concerns.

Usability: Users often face challenges when interacting with smart contracts and their transactions, particularly due to their complexity. Studies such as [6, 20, 34, 35, 51, 63, 65, 80, 84, 85, 126] highlight users' deficiency in comprehending the underlying smart contract mechanisms, leading to uninformed decisions and exposure to risks and threats. Additionally, decentralisation shifts the responsibility to users to evaluate and manage their own security in DApps [104]. These findings emphasise the critical need for clear and informative methods to aid users in understanding smart contract functionality. For example, the studies [35, 63, 131] highlight challenges faced by first-time cryptocurrency users, particularly in terms of usability and user experience. Additionally, accessibility for users with disabilities poses significant barriers, requiring exploration into potential obstacles faced by individuals with disabilities when interacting with smart contracts, which comes with an contractual enforcement [31, 64, 75].

Human-Readability. Blockchain and smart contracts introduce a novel mechanism where the code is visible to all users, allowing them to read and verify the terms embedded within the code. Therefore, the literature discusses human-readability for stakeholders who are not developers, especially in the context of legal or contractual agreements. Concerns related to human-readable contracts pertain to the comprehensibility of coding information and data for individuals with varying levels of expertise [126]. Studies emphasise that the policies and terms encoded should be readable by humans, particularly if they are legal documents [94]. It has been suggested that smart contracts should be as understandable as traditional contracts for stakeholders to grasp the written contractual agreement [33]. However, the existing programming language often lacks a straightforward mapping to natural language, impeding human understanding and reasoning [89, 133].

Governance: While it is often assumed that smart contracts operate in a decentralised manner akin to blockchains, this assumption does not accurately reflect the reality. Unlike the decentralised execution and approval of transactions inherent in blockchains, the governance of smart contracts often involves centralised mechanisms. In practice, the operation of smart contracts may be subject to centralised governance structures, including privileged accounts, third-party involvement and permission control mechanisms. The studies [62, 70, 128] collectively shed light on the centralised risks associated with smart contract governance, revealing a gap in academia concerning this area. The findings in [37, 70] specifically identifies the risk posed by access control mechanisms introducing privileges accounts into smart contracts. This risk materialises when privileged users access critical contract functionalities, potentially exposing vulnerabilities if their private keys are compromised. The access control can be a sensible measure for enhancing security in an open ecosystem; however, it may also undermine decentralisation. There is a need to balance authorisation and decentralisation within smart contract governance [70].

Cost and Fees: Fees emerge as a problematic area for users, often leading to incomplete or inaccurate understandings. The relationship between fees and transaction speed remains unclear, resulting in complexity and opacity for users [63, 74, 83, 84]. Users need to be aware of various fees, including deposit fees, exchange fees, withdrawal fees, merchant fees and network fees with their recipients. These fees are associated with different services, including wallets, exchange platforms, third-party services and miners incentives [35]. This complexity,

combined with ambiguous criteria for fee amounts and payment methods, poses challenges for users unfamiliar with blockchain technology [34] which can hinder user engagement and participation in blockchain-based applications [20, 29]. Conversely, there is an argument for hiding the intricacies of fees, particularly the gas system in Ethereum, from end-users. Exposing users to the complexities of the gas triangle—composed of gas price, gas usage, and gas limit—can lead to confusion, inefficiencies and suboptimal user experiences [83].

Trust: There is a prevailing misunderstanding that blockchain systems function entirely without trust. While smart contract code operates automatically through the blockchain’s consensus mechanism, without relying on a trusted intermediary, trust remains a significant concern. Trusting a smart contract entails placing confidence in its developers, owners and design decisions [7, 15, 35, 70]. The absence of established social contexts further complicates trust concerns surrounding smart contracts [34]. This lack of context, along with ongoing risks of centralisation, dependence on developers and manipulation, increase trust concerns regarding smart contracts and their applications such as NFTs [37, 53, 127]. Additionally, while smart contract code is known for its immutability, the recent emergence of the upgradability pattern, used to update and fix bugs in contracts, has raised several trust concerns, including the modification of functions and conditions without user awareness [55].

5.2 Current Strategies and Solutions (RQ1)

We present current solutions and interventions aimed at addressing the discussed concerns. We exclude secondary studies from the solution discussion, as they solely review the current state of the art. Furthermore, we grouped the primary studies based on their contributions into analytical research and technical solutions for better discussion. The latter category focuses on presenting methods, tools, frameworks and approaches proposed as human-centric solutions. Figure 7 summarises the categorisation of the existing solutions identified in the 77 primary studies analysed in this review.

5.2.1 Analytical Studies: This group involves the analysis of existing data, concepts, or phenomena within a specific domain, which captures the essence of examining and interpreting existing information to gain insights or draw conclusions [91]. These studies include human-based surveys, empirical investigations, conceptual and exploratory studies and comparative studies [40]. Given the infancy of this technology, these studies have provided insights and new concepts to be investigated further. Table 4 illustrates the focus areas of each analytical study in this subsection.

Human-Based Surveys: They have been conducted with users as subjects to test usability, accessibility and acceptance of decentralised applications [7, 20, 63, 104, 127], understanding trust [15], stakeholders’ interactions with smart contract wallets [131] and assess accessibility for disabled individuals [75]. Furthermore, experts have provided valuable insights into understanding user threats in cryptocurrency [34], effectiveness of development and security tools [10], evaluating societal implications of using blockchains [118] and identifying key barriers to adoption [22]. The studies [35, 80] have explained and highlighted the challenges for the Human-Computer Interaction (HCI) community to address.

Empirical Studies: These studies aim to understand specific phenomena. They investigate the impact of fees on user activity and usability [20, 29, 84], examine the trade-offs between code readability and gas consumption [116], identifying patterns of library misuse by developers [54], exploring trust and fairness issues due to NFT market manipulation [53] and testing methods to conceal gas transaction fees from end-users for the next generation of Dapps [83]. Another study provides comprehensive insights into identifying and detecting risks associated with centralised and privileged accounts [128]. Additionally, a separate study analyses the usability and security aspects of programming languages used in developing smart contracts [86].

Table 4. Overview of Analytical Studies and their Focus Areas

Type	Study	Description
Empirical Human-Based Studies	[63]	Usability testing for dApp Application
	[34]	Understanding of user-centered cryptocurrency threats
	[7]	Social and legal acceptance of end users in blockchain smart contracts for energy markets
	[118]	Evaluating design choices to support social collaborative economies
	[15]	Understanding human trust in blockchain-based systems
	[22]	Analysis to identify key barriers to adoption, the mismatch between legal requirements and IT capabilities
	[80]	Using design tools and methods, along with research and public engagement, to explain new technology from an HCI perspective.
	[75]	Analysing accessibility in crypto through qualitative data from disabled individuals
	[35]	Identifying challenges from UI to cryptocurrency-specific issues for the HCI community
	[104]	Identifying primary human security concerns and key mitigation strategies.
	[131]	Users interaction with SC wallets.
	[127]	Exploring stakeholders concerns in the NFT market.
Empirical Studies	[10]	Empirical analysis and developers survey to evaluate security tools.
	[20]	Analysing transaction fees and user acceptance using NEAR Blockchain
	[128]	Detecting centralised security risks in existing decentralised ecosystems
	[29]	The impact of fee prices on user activities on Ethereum.
	[83]	Supporting next-gen DApps that hide the gas triangle from users.
	[86]	Analysing programming practices for usability and security
	[116]	Trade-off between code readability and gas Consumption
	[84]	Analysing transaction fees and usability for Ethereum DApps.
Conceptual & Exploratory Studies	[54]	Identifying patterns of library misuse by developers.
	[53]	Large-scale data analysis of NFT transactions and marketplaces
	[70]	The dilemma between authorisation and centralisation
	[26]	Bringing attention to fundamental conceptual and methodological challenges encountered by HCI researchers
	[19]	Clarification of smart contracts in relation to the Civil Code
	[31]	An overview of blockchain technologies in the context of Industry 5.0
	[14]	Sociotechnical dynamics in SC open-source repositories.
	[9]	Exploring whether blockchain SC can reinforce the social contract between state and citizens.
Comparative Studies	[41]	Comparison of imperative and declarative languages from a legal and technical perspective.
	[12]	Usability of Obsidian programming language compared to Solidity

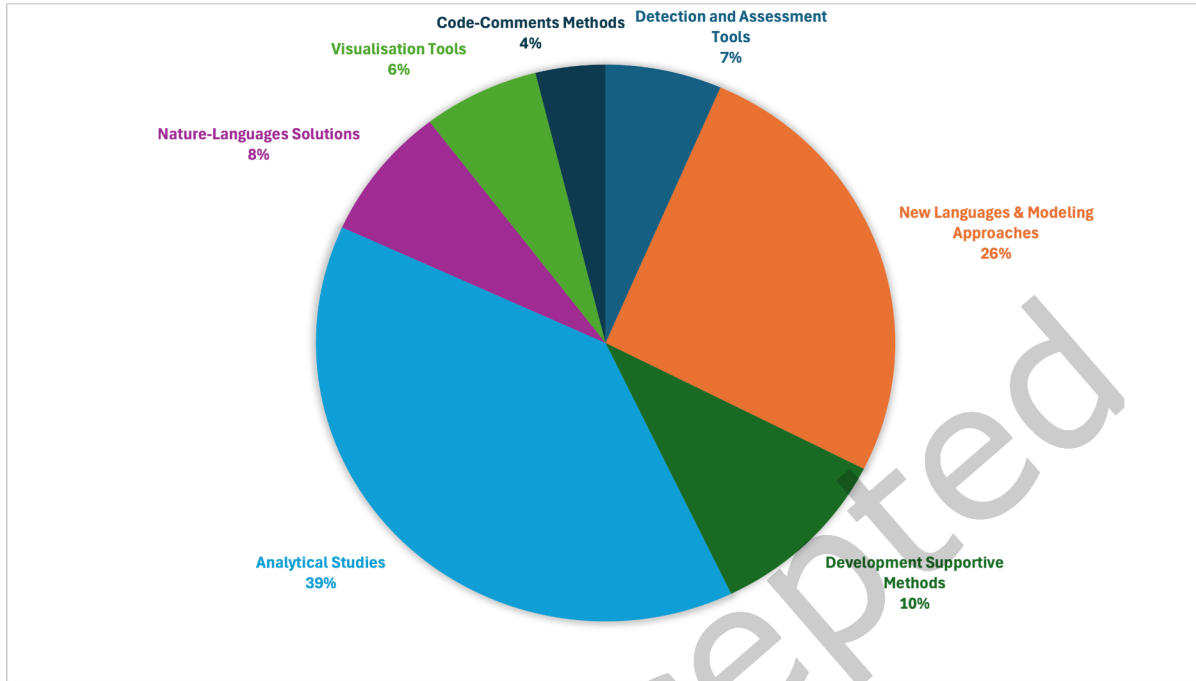


Fig. 7. Categorisation of the Existing Solutions to Address Human Concerns

Conceptual and Exploratory: The study [26] examines the intricate landscape of blockchain in terms of trust, governance, decentralisation and the fundamental challenges of HCI, whereas the study [31] explores the benefits and challenges of blockchain technologies in the context of Industry 5.0. Additionally, research has explored emerging centralisation risks in the literature, particularly issues with privileged accounts and the complexities of authorisation and security [70]. Furthermore, The study [14] analysed sociotechnical dynamics in smart contract repositories to help organisations improve development strategies. The study [9] introduces the smart social contract, using social contract philosophy to examine blockchain’s benefits and limitations in e-government systems. Lastly, the study [19] has provided valuable insights into the intersection of smart contracts and civil code, establishing a foundation for ensuring legal compliance in practical applications.

Comparative Studies: We encountered two comparative studies focusing on the languages used in smart contracts. The first study compares the differences and advantages between declarative and imperative languages [41]. It noted that while imperative languages are commonly used in practice, declarative languages offer better handling for legal and descriptive programming of smart contracts. In addition, the second study compares the Obsidian language (declarative) with Solidity (imperative) to provide further insights into their respective strengths and weaknesses [12].

5.2.2 Human-Centric Solutions. This group includes solutions and interventions such as tools, new languages, approaches, methods and frameworks. We classify and analyse these approaches based on their types, which may encompass development and interaction stages. Additionally, we specifically identify the target audiences of these solutions in our discussion.

New Programming Languages and Modelling Approaches: Addressing human considerations in smart contracts entails a focus on the languages utilised to code these contracts. This area of interest stems from the code embodying the business rules, agreements and policies, carrying obligations and enforceability upon execution. New languages, modelling approaches and their generation into executable contracts have emerged as highly reported solutions in our primary studies [6, 11, 13, 32, 43, 44, 48, 90, 94, 103, 111, 114, 125]. As a result, there is considerable interest in addressing the limitations of existing languages and how to encode policies and obligations, particularly concerning legal contracts [23, 69, 92, 93]. A few studies have also proposed requirements to be considered when designing new languages for smart contracts [2, 13, 69, 113].

Code-Comments Methods: Several studies tackle concerns regarding the readability of smart contracts through the development of comment-generation frameworks. For instance, the CCGIR framework retrieves the most similar code from the repository and reuses its comments to generate comments for smart contracts [129]. The CCGRA approach leverages retrieval knowledge to produce high-quality comments for user-defined code [133]. Moreover, the study [45] introduces a tool to detect inconsistencies between comments and code, with the aim of minimising code misuse by users and developers, and ultimately reducing the risks of vulnerabilities.

Visualisation Tools: Some studies have proposed human-centric approaches to address issues of understandability and complexity in existing languages. Visualisation techniques have been created to serve users with non-IT backgrounds, whether for drafting smart contracts or interacting with them. For instance, studies such as [76, 109, 122] advocate for visual programming platforms to create smart contracts. Visual applications featuring user-friendly interfaces have been introduced to simplify interactions with smart contracts and make them accessible to all, including socially vulnerable and underprivileged individuals [64]. A graph-based visualisation framework has also been developed for smart contracts, wallets and transaction data [65].

Nature-Languages Solutions: Researchers propose leveraging natural language techniques to enhance usability in creating and interacting with smart contracts. Studies such as [16, 33] aim to simplify smart contract creation using natural language input. Additionally, the SMARTDOC tool [51] assists users in understanding contract functions by generating natural language descriptions as user notices. Similarly, SmartBT generates readable function descriptions from smart contract bytecode to improve user understanding [126]. The study [112] introduces AI-assisted frameworks using natural language processing techniques, which provide a universal representation of contracts for a common understanding of obligations. The Tx2TXT tool automatically generates security-centric textual descriptions directly from smart contract bytecode to facilitate user decision-making before executing contracts [85].

Detection and Assessment Tools: In the pursuit of enhancing human-centric aspects of smart contracts, various tools have been developed to address readability, fairness and ownership concerns. The FairCon framework automatically verifies the fairness properties of smart contracts [73]. Also, the study [55] presents a bytecode-based classification model to detect contracts using the proxy upgrade pattern, aiming to reduce trust issues. The Ethpector is a technical solution to automatically extract privileged parties from smart contract bytecode [37]. Furthermore, FunRedis is a tool that reduces gas fees by prioritising frequently used functions in smart contract bytecode at runtime [74]. Finally, the study [8] introduces a tool to assess code readability automatically.

Development Supportive Methods: Several approaches have been proposed to support developers and address shortcomings in smart contract development. These methods aim to streamline contract creation and provide flexibility for modification. For instance, one approach employs model-driven engineering to streamline smart contract design and development [42], while another method facilitates their collaborative creation using supportive drafting tools [89]. The study [81] introduces a novel framework for secure oracle development. The ConSol is a Solidity extension enabling developers to specify and monitor behavioral contracts for functions and

addresses [121]. The Giffar method [99] enables dynamic generation of smart contract code while a decentralised application remains operational. Additionally, The Long Short-Term Memory Recurrent Neural Networks (LSTM-RNN) generates contract templates [46]. When addressing legal modification concerns, both [66] and [77] propose solutions—a system architecture for modification and a set of design standards, respectively. Lastly, to tackle privileged accounts and centralisation issues, the study by [62] offers a library for responsible ownership and management of ERC20 tokens.

5.3 Mapping Human Concerns with Quality Attributes (RQ2)

In this section, our objective is twofold: first, to comprehend the common qualities associated with human concerns as discussed in the primary studies, and mapping these qualities with the terminologies outlined in the ISO/IEC/IEEE 24765 [57] standard. Secondly, we adopt NIST [61] trustworthiness characteristics to identify new qualities that may have been overlooked in the existing literature. The method and rationale for choosing these standards are described in Section 3.6.

5.3.1 Common Qualities. Table 5 presents the most commonly reported qualities and their mapped definitions from the ISO/IEC/IEEE 24765 standards [57]. We observed terms and characteristics in the context and mapped them with the most appropriate qualities defined in [57]. The second column of Table 5 provides representative examples of the terms used to describe concerns, which we then mapped with the terminologies in the first column. Additionally, the table includes a sample of primary studies where specific terms and characteristics of smart contracts were identified. The most commonly reported quality attributes with concerns across both phases of development and interaction include understandability, transparency and accountability, simplicity and learnability, usability and accessibility, safety and fairness. The selected primary studies have also reported other quality attributes, such as flexibility, readability and efficiency.

5.3.2 Overlooked Qualities. Smart contracts, as a groundbreaking technology, necessitate thorough exploration as they are still in their infancy and lack standardisation and guidelines to address the challenges encountered by human in the loop. Current standards fail to adequately address the unique characteristics of immutability, automation, and enforcement inherent in smart contracts. Therefore, we extend our examination of smart contracts by exploring trustworthiness within systems that automate decision-making such as AI. Although smart contracts primarily adhere to more straightforward if/else statements for decision-making, their complexity arises during development and execution due to the immutability of blockchain, where decisions are final and irreversible, and the enforceability of outcomes in smart contracts may entail financial obligations.

Attributes of trustworthiness are deeply connected to behaviours in social and organisational settings. They are shaped by the decisions of those who develop these attributes and by interactions with individuals who offer insights and oversight to these systems [61]. Therefore, the concept of trustworthiness is akin to human considerations in designing and interacting with systems, which can help us explore potential qualities that can enhance the design and development of smart contracts.

The NIST standards [61] outline characteristics and principles for AI trustworthiness and they define quality attributes using well-known ISO standards. We leverage these characteristics to systematically identify qualities that may have been overlooked or not previously considered in the design of smart contracts. The NIST delineates seven key characteristics of trustworthiness: (1) Valid and Reliable, (2) Safe, (3) Secure and Resilient, (4) Accountable and Transparent, (5) Explainable and Interpretable, (6) Privacy-Enhanced, and (7) Fair – with Harmful Bias Managed, as presented in Figure 8. The characteristics of "Valid and Reliable" serve as the foundation for the other trustworthiness attributes, while "Accountable and Transparent" are interconnected with all other characteristics, as illustrated in the vertical box. However, it is important to note that trade-offs typically occur.

Table 5. Mapping of Most Discussed Quality Attributes (RQ2)

Standard Mapping	Representative Terms	Sample
Understandability	"understanding SCs before executing", "deep technical understanding", "must understand the contents", "poorer understandability", "struggle to understand", "cannot understand functionality", "interpreting their meaning"	[6, 8, 51, 89, 112]
Transparency & Accountability	"does not speak to concrete implementations", "threat to accountability", "party's power", "trust in individuals/institutions", "destroying SCs", "control operation", "violate decentralization", "underlying intent", "parties without knowledge"	[26, 37, 65, 85, 128]
Simplicity & Learnability	"familiarity of the developers", "not easy to implement", "distinct terminologies", "steep learning curve"	[12, 42, 46, 76]
Usability & Accessibility	"how interfaces prevent adoption", "user interfaces suffer", "friendly to lawyers", "for the socially vulnerable", "accessibility violations", "multi language issue arises"	[2, 35, 63, 75, 89]
Compliance	"legally-binding DAO", "regulation and legislation uncertainty", "contracts require signatures", "conflict with SCs"	[7, 19, 22, 23, 113]
Safety	"protection", "stronger safety properties", "loss of money", "safety concern"	[16, 103, 111]
Fairness	"fair, secure, flexible", "biases will be encoded", "is unfair to certain participants"	[7, 26, 73]

Not all characteristics apply equally in every setting; some may be more or less important depending on the situation [61].

Given the broader discussion of smart contract concerns, including technical and human aspects, principles such as validity, reliability, safety, security, privacy and fairness are already being discussed. However, characteristics that tend to be overlooked based on the seven principles are *explainability* and *interpretability*. Explainability involves representing the mechanisms underlying the operation of systems, while interpretability pertains to understanding the significance of system outputs within their intended functional contexts. Collectively, explainability and interpretability aid system operators and users in gaining deeper insights into the functionality and reliability of the system, including its outputs [61]. Additionally, smart contracts have had limited in-depth discussions on accountability and transparency, as emphasised in the human concerns Section 5.1. It is often assumed that smart contracts are transparent, similar to blockchain technology. However, this misconception has led users to trust risky smart contracts, resulting in financial losses [56]. The decision-making processes, policies, terms and privileged controls within smart contracts are not typically transparent nor explainable. This lack of transparency also impacts the accountability of these decisions.

Therefore, we recognise that the human-centred attributes of explainability and interpretability are frequently neglected in the domain of smart contracts. Additionally, the aspects of transparency and accountability have not been thoroughly explored in existing research. These attributes are essential for the design of human-centric

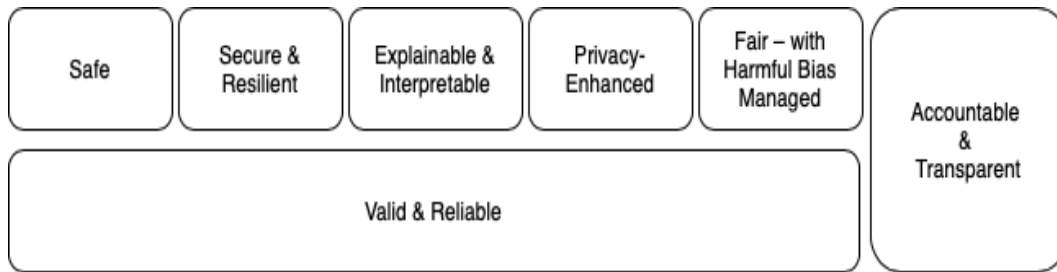


Fig. 8. The AI Trustworthiness Characteristic by NIST [61]

smart contracts, as they address critical questions such as "how decisions are made," "who is responsible or involved in making those decisions," and "why specific actions are taken."

6 Discussion

This section reflects on the key findings of our review and highlights critical gaps and opportunities for advancing human-centric smart contract research. The discussion is structured around insights into current research practices, the lack of human considerations in smart contract design, and directions for future work. We also acknowledge the potential validity threats of this study and the measures taken to mitigate them.

6.1 Current Insights and Future Directions

Based on our findings, the literature on human considerations has primarily focused on developing new programming languages, code generation approaches and external tools while overlooking aspects related to the design of smart contracts. Several gaps can steer future directions as follows:

Our review reveals that most human-related research in the context of smart contracts has concentrated on programming languages, code generation approaches, and external tools aimed at improving understandability and usability. However, limited attention has been given to the requirements and design stages of smart contracts themselves. Recent efforts have focused on establishing new languages and tools to create smart contracts, as presented in Section 5.2. According to [117], more than 100 new languages have been proposed for smart contracts. This proliferation of languages could complicate development efforts and present challenges for developers. Future efforts should focus on designing smart contracts with tailored quality attributes that reflect their unique nature, including the development of new definitions and standards for their characteristics and properties.

Furthermore, many of these language and modelling innovations are driven by legal and compliance concerns. For instance, the conflict between smart contracts and data protection regulations such as the GDPR—particularly the "right to be forgotten"—has received considerable attention [7, 22, 39, 88]. However, less emphasis has been placed on Article 22 of the GDPR, which addresses individuals' rights to receive meaningful information about automated decision-making processes [95]. This article is especially relevant to smart contracts, which autonomously execute and enforce decisions without direct human intervention. As noted by [30], smart contracts may qualify as fully automated processes, making the right to explanation a critical yet underexplored consideration in this domain.

While much of the literature—and our review—focuses on Ethereum-based implementations, it is important to acknowledge that smart contracts are not limited to a single platform. Alternative architectures, such as Hyperledger Fabric (where smart contracts are referred to as chaincode) and private blockchain environments that do not involve cryptocurrencies, offer distinct operational and design models. These differences can significantly influence how human-centric attributes are interpreted and applied. Incorporating a broader view of platform

diversity in future research may uncover platform-specific challenges and opportunities for enhancing trust, usability and compliance in smart contract systems.

Alongside platform-related considerations, another critical dimension concerns the interaction between users and smart contracts. Despite the growing interest in improving interaction with smart contracts through visualisation tools and code generation techniques [51, 64, 76, 85, 109, 122], the literature lacks dedicated methodologies that incorporate human needs and perspectives into the design phase. As smart contracts become increasingly complex, relying solely on external tools and approaches may prove insufficient and overwhelming for users who attempting to navigate their usage effectively.

In parallel, an emerging trend in the blockchain space involves the generation of smart contract code using AI tools powered by large language models (LLMs) [18]. This development presents new opportunities for research into how such automatically generated contracts can be aligned with human-centred principles. These tools can streamline development, but they can also obscure the logic behind decisions and introduce hidden risks. Therefore, this emerging area underscores the importance of exploring the explainability in both AI-generated output and smart contract decision-making processes.

Human concerns also revolve around understandability, transparency, accountability, usability, accessibility, safety and fairness, all of which are integral characteristics for ensuring the trustworthiness of the system. However, our findings indicate that explainability and interpretability remain significantly underexplored. Trust in decentralised systems depends not only on the outcomes of contract execution but also on clarity regarding how decisions are made, who is responsible, and what inputs are involved. Another key gap lies in the reasoning and justification processes within smart contract decisions. While certain behaviours (e.g., token swaps or auction mechanisms) have been explained in prior work [85], users often face unexpected outcomes due to opaque logic. This indicates the need for mechanisms that provide clear, accessible explanations of how decisions are made, especially to prevent misuse or manipulation by privileged actors. Access to this information is necessary to support meaningful understandability, transparency, accountability, and ultimately, explainability.

Although these future directions have been discussed throughout this section, we summarise them here to offer a concise roadmap for advancing human-centred smart contract research:

- *Human-Centric Smart Contract Design:* Future research should focus on designing smart contracts with tailored quality attributes that reflect their unique nature. This requires the development of new definitions and standards for smart contract characteristics and properties to better design human-centric smart contracts.
- *Examining Web3-Based Interaction in Smart Contract:* The impact of Web3 technologies on human-centric smart contract design has received limited attention in the academic literature, despite their increasing adoption in industry to support more interactive and user-friendly decentralised applications. Future research should explore how Web3-based interaction models can be designed to align with usability, trust, and accessibility principles, thereby supporting more inclusive and intuitive smart contract experiences.
- *Right to Receive Meaningful Information:* Aligning smart contract design with GDPR Article 22 warrants further investigation. Research should explore how to communicate decision logic, responsible entities, and execution conditions in a manner that preserves decentralisation while enabling compliance.
- *Enhancing Explainability and Interpretability:* Future research can focus on embedding explainability and interpretability into the core of smart contract design. This approach can include developing mechanisms that make decision logic transparent, enable users to follow execution flows, and detect or mitigate potential manipulation—particularly in scenarios involving privileged access or automated actions.
- *Gaps in Smart Contract Development:* There is a pressing need for standardised methodologies and tools to support developers in building human-centric smart contracts. Existing tools often fall short, particularly in enhancing code readability, learnability, simplicity and maintainability. Addressing these challenges

effectively requires interdisciplinary research that brings together human-computer interaction (HCI) principles and blockchain engineering, ensuring that developer tools and smart contract design both align with human-centred requirements.

- *Smart Contracts Across Platforms:* The literature predominantly focuses on Ethereum-based implementations; however, both alternative public and private blockchain platforms present distinct architectural and operational characteristics. These differences may significantly affect how human-centric attributes—such as usability, trust, and compliance—are interpreted and applied. Future research should investigate how platform-specific constraints and affordances shape the integration of human-centric principles in the design, development and interaction of smart contracts.

6.2 Threats to Validity

This section outlines the potential threats to validity identified in our study, guided by the insights provided in [4, 124].

6.2.1 Internal validity: To mitigate the impact of irrelevant variables and potential biases in our study, we established a rigorous research protocol following guidelines by Kitchenham et al. [68]. Initially, we conducted a scoping review to formulate the search string by experimenting with a few databases. We then implemented a rigorous selection strategy with explicit inclusion and exclusion criteria. To enhance our coverage, we employed forward and backward snowballing techniques [123], mitigating the risks of missed studies by automated searches. Additionally, authors independently conducted the paper selection process and resolved discrepancies through group discussions. To further minimise biases, we ensured all authors shared a common understanding and aligned the data extraction process with the research questions.

6.2.2 Construct validity: A potential issue arises when the operational definitions or measurements of constructs (that is, concepts or variables) in a study do not accurately represent the theoretical concepts intended to measure. Given the broad nature of concerns and human aspects, which could potentially introduce conflicting concepts during data extraction and synthesis, we carefully defined the terminology of concerns and perspectives related to humans. Specifically, we used well-established standards to define these terminologies and implemented a systematic approach to ensure consistency and clarity. Additionally, we considered secondary studies to cover insights not addressed in the primary studies and assess our identified concerns' coverage.

Additionally, the term "human-centric" refers to the explicit consideration of human needs, perspectives, and interactions throughout the lifecycle of smart contracts—from design and development to deployment and use. Rather than viewing smart contracts solely as technical artefacts, a human-centric approach recognises the diverse roles and expectations of stakeholders, including developers, end-users, regulators and decision-makers. By framing our analysis through this lens, we aim to uncover how existing literature addresses—or overlooks—the human dimension in smart contract systems and what qualities are necessary to support their broader adoption and trustworthiness. On the other hand, "human involvement", as mentioned in the reviewed studies, refers to the extent of direct human participation in the operation or maintenance of smart contracts, such as centralised privileges. Although blockchain systems inherently promote decentralisation and automation to minimise human intervention, many cases in studies highlight the ongoing need for human decisions and operations. The distinction between these terms has been clearly discussed in our analysis.

6.2.3 Conclusion validity: A potential threat to conclusion validity is the possibility of incomplete coverage of smart contract concerns and classifications, as other classifications or themes may exist. To mitigate this risk, we employed thematic analysis [17], an iterative process that allowed us to refine our classifications as new concepts emerged. We also consulted secondary studies and standards to ensure any missing classifications or concerns

were noticed. Nevertheless, our classification system remains adaptable and capable of evolving to accommodate new additions and changes over time.

7 Related Work

In this section, we first reviewed related SLRs in the blockchain smart contract domain to understand the existing body of knowledge and prior secondary research. Building on this, we then examined the classification of concerns in smart contract research, analysing how various aspects have been categorised across different studies. This approach enabled us to evaluate existing classifications and groupings of concepts within our research framework while developing a comprehensive understanding of the evolving landscape of smart contracts.

Existing SLRs on smart contracts can be categorised into three main groups: (1) technical reviews, (2) user-centric reviews, and (3) scope-focused reviews. Although these studies provide valuable information, our work differs by addressing smart contracts from a human-centred perspective, emphasising usability, trust, and sociotechnical interactions. Table 6 summarises the key studies in each category.

Technical Reviews: Alharby et al. [3] provided insight into the current research landscape, classifying studies into six categories: security, privacy, software engineering, application, performance and scalability. They have noted a lack of research on scalability, particularly in executing contracts in parallel to enhance throughput. Additionally, Wang et al. [120] presented a framework for smart contracts with six layers: infrastructures, contracts, operations, intelligence, manifestations and applications, highlighting security vulnerabilities. Khan et al. [67] categorised existing smart contract studies into two categories: smart contract improvement and smart contract usage. The former addresses challenges such as functionality verification, performance optimisation, vulnerability mitigation and trustworthy data feeding. The latter focuses on domain-specific challenges through smart contract utilisation. Moreover, Taherdoost [108] have highlighted challenges such as the lack of solid data processing capacity, effective smart contract management and security vulnerabilities. Ante [5] conducted a bibliometric analysis of smart contracts research, identifying several main research streams: technical foundations, blockchain applications for IoT, standardisation, verification and security. It highlights emerging clusters such as smart contracts and the law, indicating their interdisciplinary nature. Despite various findings, the study underscores the uncertainty surrounding smart contracts' potential.

These reviews primarily focus on the technical challenges associated with smart contracts. In contrast, our study takes a different approach by addressing challenges from a human perspective. We view smart contracts as social mechanisms, emphasising the importance of human understanding, trust and interaction in their design.

User-Centric Reviews: Dixit et al. [21] conducted a systematic literature review on smart contract automation models which focuses on technical features and legal significance. They highlight that existing approaches primarily cater to technical users, limiting accessibility for non-technical users and neglecting the social aspects. Our study extends this work by incorporating additional user perspectives, contributing to a broader understanding of smart contract usability. Additionally, the review by Fröhlich et al. [36] focuses on the interaction phase for a single use case, cryptocurrency, whereas our study provides a broader perspective. They identified six themes: trust, motivation, wallet usability, user engagement, application-specific use cases and support tools. Their review emphasises the importance of trust in decentralised systems and advocates for sociotechnical design perspectives. It aids in understanding blockchain interaction design and suggests future HCI research directions.

Scope-Focused Reviews: Vacca et al. [115] noted that reviews of the existing literature focus mainly on security and biomedical applications, rather than software engineering. Most of the research centres on Ethereum, with limited attention to Bitcoin and Hyperledger. This gap underscores the need to investigate software engineering issues in different blockchain platforms. The study highlights key challenges such as integrating blockchain with existing systems and evaluating associated costs and benefits. Another systematic review, conducted by Haque et al. [47], synthesised previous work on GDPR-compliant blockchains. They highlighted key GDPR articles,

Table 6. Related Work and Their Focus Areas

Study	Focus Area
[3]	Classifies technical challenges into six main categories: security, privacy, software engineering, application, performance and scalability.
[5]	Identifies main research streams, covering technical foundations, blockchain applications for IoT, standardisation, verification and security
[21]	Addresses automation and generation of smart contracts from a user perspective.
[36]	Cryptocurrency human challenges classified into six themes: trust, motivation, usability, user engagement, application-specific use cases and support tools.
[47]	Highlights key GDPR articles issues related to blockchain and smart contracts.
[67]	Categorises challenges into two primary categories: improvement and usage.
[108]	Discusses various technical and management challenges.
[115]	Focuses on challenges in software engineering aspects and platforms other than Ethereum.
[120]	Highlights general challenges of smart contracts across system layers.

particularly issues with data deletion and modification. The study also explored the distribution of roles among actors, emphasising GDPR compliance in the IoT and blockchain-based industrial data contexts. However, it lacked discussion on Article 22, which pertains to providing meaningful information for automated processes and is linked to the informal right to explanation.

7.1 Classification Considerations

In addition to reviewing the primary scope of existing SLRs, we extended our analysis to include the classifications used in a broader set of secondary studies—such as surveys and literature reviews—identified during the paper selection process. This extension aimed to validate our classification framework and identify any relevant insights that may have been overlooked in the primary studies.

These secondary studies cover a wide range of topics, including overview of smart contracts in general [3, 67, 97, 108, 115, 120], legal perspectives [21, 25, 38], smart contract languages [24, 117], user interaction and social dimensions [21, 36], and ethical considerations [110]. Table 7 presents the thematic classifications derived from the reviewed secondary studies alongside those identified in this study.

Apart from addressing technical aspects such as security and scalability, these secondary studies highlight several human-centred concerns that are consistent with those identified in our analysis. One frequently mentioned concern is privacy. However, we chose not to include privacy in our classification, as it is more closely associated with the fundamental characteristics of blockchain technology than with the design or interaction of smart contracts. Given blockchain's inherent transparency—where transaction data is publicly accessible—privacy challenges primarily arise at the infrastructure level. While privacy is undeniably an important human concern in the broader blockchain ecosystem, it falls outside the specific scope of this study, which focuses on human-related issues within the context of smart contract design and use.

8 Conclusion

In this article, we present a systematic literature review to explore human concerns in blockchain smart contracts and investigate current human-centric approaches and methods. We selected 77 primary studies following a systematic search strategy. We found that human concerns in smart contracts are addressed in the literature surrounding two main stages: development and interactions, from the perspective of stakeholders. We classify development concerns into three categories: *language*, *legality*, and *ethical and social implications*. Language was the most frequently reported concern in our collection, highlighting issues related to complexity, code

Table 7. Concerns Classification of Secondary Studies

No	Study	Categorization Description	Classifications
1	[67]	Survey of smart contracts from technical and application perspectives.	Improvement (model-driven, optimization-driven), Usage (resource-driven, cross-organizational)
2	[120]	Concerns based on the six-layer architecture for smart contracts.	(1) Operational & Technological Developments, (2) Challenges & Progress (Contract Vulnerabilities, Blockchain Limitations, Privacy, Legal Issues), (3) Application Scenarios
3	[108]	Current smart contract applications and their transformative effects.	Processing, Acceptance, Immutability, Integrity, Usability, Security, Legal Issues, Privacy
4	[97]	Develops a research framework for blockchain studies.	(1) Technological Developments (Design & Features), (2) Limitations (Neglecting Application, Value Creation, Governance)
5	[3]	Smart contract development concerns.	Security, Privacy, Software Engineering, Application, Performance, Scalability
6	[115]	Challenges in smart contract engineering.	Smart contract testing, Code analysis, Metrics, Security, DApp performance, Blockchain applications
7	[21]	Automation models based on technical and legal perspectives.	Constraints (Semantic rules, Meta rules), Accessibility (Human readability, UX), Regulatory transparency (Legal prose, Denotational semantics)
8	[36]	Human interaction in cryptocurrency and blockchain.	(1) Trust, (2) Motivation & Risk Perception, (3) Wallet Usability, (4) User Engagement, (5) Blockchain Applications
9	[25]	Analyses smart contracts from a legal perspective.	Contracting environments, Enforcement mechanisms, Trust, Efficiency, Regulatory responses
10	[38]	Examines legal intent, remedial issues, and enforcement.	Capacity, Contracting under mistake, Legal intent, Certainty, Content interpretation, Security, Scalability, Workforce impact
11	[24]	Reviews existing smart contract languages.	Semantic suitability, Workflow suitability, Expressiveness
12	[117]	Reviews state-of-the-art research on smart contract languages.	Verification, Financial contracts, Security, Privacy, Safety, Legal agreements, Business processes, Formalization, Natural language
13	[110]	Discusses blockchain ethics and decentralization.	Technology stack, Cryptocurrencies, Smart contracts, Decentralization
14	This study	Analyzes expressed human concerns in smart contract development and interaction and their existing interventions.	Concerns Development (Language: Complexity, Code-readability, Expressiveness; Legality, Ethical and Social Concerns), Interaction (Usability, Human-readability, Trust, Governance, Cost and Fees). Existing solutions are classified into Analytical Studies and Human-Centric Solutions

readability and expressiveness. These concerns impact technical developers, non-technical or domain experts and collaborative development teams. On the other hand, interaction concerns, from the perspective of end-users, revolved around *usability, human readability, governance, trust, and cost and fees*.

Based on the findings, we observed that most research has focused on programming languages, their modelling approaches and the use of external tools to enhance understandability and usability. However, there is a noticeable lack of attention to the requirements and design aspects of smart contracts. In reviewing the literature, we identified a range of quality attributes related to human-centred concerns. The most commonly reported attributes include understandability, transparency, accountability, simplicity and the learnability of programming languages, along with usability, accessibility, compliance, safety and fairness. In particular, explainability and interpretability appear to be largely overlooked, while transparency and accountability have received only limited attention.

These findings point to promising opportunities for developing new techniques and frameworks that enhance the trustworthiness of smart contracts. Future research should focus on designing contracts with quality attributes aligned with their decentralised nature, and on creating methods that clarify decision-making processes to support explainability and accountability.

In conclusion, this review contributes a focused and systematic synthesis of human concerns in blockchain smart contracts. Its strengths are rooted in the structured selection and analysis of peer-reviewed studies, the explicit focus on stakeholder perspectives and the identification of underexplored qualities such as explainability and interpretability. At the same time, the review is shaped by specific design choices, such as limiting the scope to peer-reviewed literature published up to 2024 and focusing primarily on public blockchain platforms such as Ethereum. While these parameters provide depth and rigour, they may also omit emerging insights from alternative sources such as grey literature or more recent developments. We expect that future reviews will expand on this foundation by integrating a broader range of sources and perspectives to enhance the understanding of human-centric smart contract design.

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A Appendix

Table 8 illustrates the quality assessment criteria and scores for each primary study selected for the systematic literature review.

B SLR Primary Studies Summary

Table B provides a summary of the primary studies included in the systematic literature review.

Table 8. Quality Assessment Results

Study	Rationality			Rigor			Credibility		Total	Study	Rationality			Rigor			Credibility		Total
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8			Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	
[2]	1	1	1	1	0.5	0.5	0.5	5.5	6	[6]	1	1	1	1	1	0.5	0.5	0	6
[7]	1	1	1	1	1	1	1	1	8	[8]	1	1	1	1	1	1	1	0	7
[9]	1	1	1	1	1	1	1	1	8	[10]	1	1	1	1	1	1	1	1	8
[11]	1	0.5	1	1	0.5	0.5	1	0	5.5	[12]	1	0.5	0.5	1	1	1	1	1	7
[13]	1	1	1	0.5	0.5	0	0.5	0	4.5	[14]	1	1	1	1	1	1	1	1	8
[15]	1	1	1	1	1	1	1	0	7	[16]	0.5	0.5	1	1	1	1	1	0	6
[19]	1	1	1	1	1	0.5	1	0	6.5	[20]	1	1	1	1	1	0.5	0.5	0.5	6.5
[22]	1	1	1	1	1	1	1	0	7	[23]	1	1	1	1	1	1	1	0	7
[26]	1	1	1	1	1	1	1	0	7	[29]	1	1	1	1	1	1	1	0.5	7.5
[31]	1	1	1	1	1	1	1	0	7	[32]	1	1	1	1	1	1	1	0	7
[33]	1	1	1	1	0	0	1	0	5	[34]	1	1	1	1	1	1	1	0	7
[35]	1	1	1	1	1	1	1	0	7	[37]	1	1	1	1	1	1	1	1	8
[41]	1	1	1	1	1	1	1	0	7	[42]	1	1	1	1	0.5	0.5	1	1	7
[43]	1	1	1	1	1	1	1	1	8	[44]	1	1	1	1	1	1	1	0.5	7.5
[45]	1	1	1	1	1	1	1	1	8	[46]	1	1	1	1	1	1	1	0	7
[48]	1	1	1	1	1	1	0.5	0	6.5	[51]	1	1	1	1	1	1	1	1	8
[53]	1	1	1	1	1	1	1	0	7	[54]	1	1	1	1	1	1	1	1	8
[55]	1	1	1	1	1	1	1	0	7	[62]	1	1	1	1	1	1	1	0	7
[63]	1	1	1	1	1	1	1	0	7	[64]	1	1	1	1	0.5	0	0.5	0	5
[65]	1	1	1	1	1	1	1	0	7	[66]	1	1	1	1	1	1	1	0	7
[69]	1	1	1	1	0.5	0.5	1	0.5	6.5	[70]	1	1	1	0.5	0.5	0	0.5	0	4.5
[73]	1	1	1	1	1	1	1	1	8	[74]	1	1	1	1	1	1	1	0	7
[75]	1	1	1	1	1	1	1	1	8	[76]	1	1	1	1	1	1	1	0	7
[77]	1	1	1	1	1	1	1	0	7	[80]	1	1	1	1	1	1	1	0	7
[81]	1	1	1	1	0.5	0.5	1	0	6	[83]	1	1	1	1	1	1	1	0	7
[84]	1	1	1	1	0.5	1	1	0	6.5	[85]	1	1	1	1	1	1	1	0	7
[86]	1	1	1	1	1	1	1	1	8	[89]	1	1	1	1	1	1	1	0	7
[90]	1	1	0.5	1	0.5	0.5	1	0	5.5	[92]	1	1	1	1	1	1	1	1	8
[93]	1	1	1	1	1	1	1	1	8	[94]	1	1	1	1	1	1	1	0	7
[99]	1	1	1	1	0.5	1	1	1	7.5	[103]	1	1	1	1	1	1	1	0	7
[104]	1	1	1	1	1	1	1	0	7	[109]	1	1	1	1	0.5	0.5	0.5	0	5.5
[111]	1	1	0.5	1	1	1	1	0	6.5	[112]	1	1	1	1	1	1	1	0	7
[113]	1	1	0.5	1	0	0.5	1	0	5	[114]	1	1	1	1	0	0	0.5	0	4.5
[116]	1	1	1	1	1	1	1	1	8	[118]	1	1	1	1	1	1	1	0	7
[121]	1	1	1	1	1	1	1	1	8	[122]	1	1	1	1	0.5	1	1	0	6.5
[125]	1	1	1	1	1	1	1	0	7	[126]	1	1	1	1	1	1	1	0	7
[127]	1	1	1	1	1	1	1	1	8	[128]	1	1	1	1	1	1	1	0	7
[129]	1	1	1	1	1	1	1	1	8	[131]	1	1	1	1	1	1	1	1	8
[133]	1	0.5	0.5	1	1	1	1	1	7										

Table 9. List of Primary Studies with Synthesis of the Results

No	Study	Concern	Interventions	Stakeholders	Area
1	[42]	Designing and creating SC has a steep learning curve, posing a challenge.	Using Model-Driven Engineering to aid design and development with Domain Specific Language	Developers (IT & non-IT)	Language
2	[99]	Complexity of smart contracts' construction	JSON-based module for contract modeling, and enables automatic code generation with TypeScript.	Developers (IT)	Language
3	[114]	Meeting stakeholders' goals and policies as complexity increases	Automates contract generation using goal models for tasks and policies.	Developers (IT)	Language
4	[46]	Complex coding process leads to unnormalised code, causing development and maintenance issues.	Uses Long Short-Term Memory Recurrent Neural Networks (LSTM-RNN) to generate contract templates and streamline coding	Developers (IT)	Language
5	[11]	Translating business rules into smart contracts is challenging due to frequent reuse across contracts	Domain-specific ontologies and semantic rules.	Developers (IT & non-IT)	Language
6	[13]	Languages should prioritise user needs, detect critical bugs at compile time, and be blockchain-agnostic	New Language "Obsidian" which aligns with these requirements	Developers (IT)	Language
7	[125]	Disconnect between contractual clauses and code hinders understanding and efficiency.	High-level domain-specific language auto-transformable to implementation	Developers (IT & non-IT) & Users	Language
8	[111]	Development is labor-intensive and error-prone, requiring business collaboration.	Human-readable documents using DSL4SC templates and parameter mapping	Developers (IT)	Language
9	[86]	Existing languages require a steep learning curve and lead to bugs.	Empirical evaluation of existing languages for usability and security.	Developers (IT & non-IT) & Users	Language
10	[16]	SC lacks social interaction capabilities and overlook non-digital currency applications.	Fides, a natural language contract creation framework.	Developers (IT & non-IT)	Language
11	[12]	Challenges in writing safe smart contracts.	Compares usability in Obsidian vs. Solidity.	Developers (IT)	Language
12	[90]	Lack of unified standards complicates development and automatic generation	Auto-generate SC using UML and knowledge extraction.	Developers (IT)	Language
13	[121]	Validation approaches fail to help developers define interfaces, especially for opaque contracts.	ConSol: a Solidity extension for specifying and monitoring behavioral contracts.	Developers (IT)	Language

No	Study	Concern	Interventions	Stakeholders	Area
14 - 15	[43, 44]	Development issues due to diverse blockchain platforms with unique syntax	DSML language for deploying smart contracts on multiple blockchains	Developers (IT)	Language
16	[109]	Solidity requires programming skills, challenging non-programmers.	Visually creates Ethereum SC with a direct-manipulation interface.	Developers (IT & non-IT)	Language
17	[76]	Solidity's complexity makes it hard for average users to design SC.	Visual programming platform for users	Developers (non-IT) & Users	Language
18	[48]	Enabling domain experts with non-IT backgrounds to collaboratively understand, discuss, and specify the contract.	A specification language to facilitate collaborative design	Developers (IT & non-IT) & Users	Language
19	[112]	Semantic consistency and consent among diverse participants.	Automate converting contract clauses into MNL sentences, understandable by both smart contracts and humans.	Developers (IT & non-IT) & Users	Language
20	[116]	Readability vs gas consumption	Empirical study - Trade off analysis of code readability and gas consumption	Developers (IT)	Language
21	[8]	Lower readability makes smart contracts hard to understand and reuse	iSCREAM, a tool to help developers and researchers estimate code readability	Developers (IT)	Language
22	[129]	Limited budgets and experience result in missing or inaccurate code comments	Automatic code comment generation method	Developers (IT)	Language
23	[45]	Comment-code inconsistencies can mislead developers and introduce vulnerabilities	Detecting comment-code inconsistencies	Developers (IT)	Language
24	[133]	Lack of effective comments makes understanding SC challenging.	Uses retrieval knowledge to generate Solidity code comments.	Developers (IT & non-IT) & Users	Language/ Human-Readability
25	[32]	Limited mechanisms exist to make specification and interpretation accessible to a broader audience.	Model supports semi-automated translation of human-readable contracts	Developers (IT)	Language
26	[122]	Creation is inaccessible to non-technical experts.	Graphical programming language using modularised legal contracts.	Developers (IT & non-IT)	Language
27	[6]	Lack of code expressiveness hinders user understanding	MDA approach to enhance trust and clarity	Developers (IT) & Users	Language/ Usability
28	[89]	Contracts are not human-readable, hard to modify, and hinder collaborative drafting and mutual consent."	Intelligible Description Language Contract (IDLC) enables collaborative drafting like a text editor.	Developers (IT & non-IT) & Users	Language/ Human-Readability

No	Study	Concern	Interventions	Stakeholders	Area
29	[41]	Declarative vs. imperative languages	Comparison Analysis	Developers (IT)	Language
30	[94]	Existing language lacks clear mapping to natural language, hindering human understanding	Design concepts & domain-specific language (DSL)	Developers (IT)	Language/ Human- Readability
31	[23]	Developing legally-binding DAOs presents a complex challenge	SLCML: Markup language for legally-binding DAOs.	Developers (IT & non-IT)	Language
32	[2]	Trust concerns in development	Requirements for a lawyer-friendly, human and machine-readable contract authoring language	Developers (IT & non-IT)	Language
33	[92]	Knowledge gap between developers and legal experts	Symboleo: Formal specification language for legal contracts.	Developers (IT & non-IT)	Language
34	[93]	Legal experts cannot directly encode contract requirements into SC.	Symboleo2SC: a tool that automates SC code generation from legal specifications.	Developers (non-IT)	Language
35	[103]	Trade-off between expressiveness and safety in language design.	Scilla: Intermediate-level language for safe smart contracts	Developers (IT)	Language
36	[54]	Libraries misuse by developers due to lack of security awareness.	An empirical study to identify patterns of library misuse	Developers (IT)	Language
37	[10]	Security tools fail to meet developers' and auditors' needs, risking DeFi users.	Empirical and human based survey to evaluate security tools and their effectiveness	Developers (IT)	Language
38	[74]	Lack of gas mechanism understanding and poor optimisation raise costs for developers and users.	FunRedis: Tool to reduce gas fees by prioritising frequently used functions in SC bytecode.	Developers (IT) & Users	Language /Cost
39	[113]	Challenges stem from design approach, not programming languages used.	Beagle framework integrates law into smart contracts	Developers (IT)	Language/ Legality
40	[69]	Developing and verifying legal smart contracts challenges	Unified model for language requirements.	Developers (IT)	Language/ Legality
41	[22]	Conflicts with existing laws, limitations at the individual contract level, and current technical design issues	Human-based survey - Identifying key barriers to adoption	Developers (IT & non-IT) & Users	Legality
42	[77]	Traditional legal tools fail in altering and updating SC	Standards for altering and undoing smart contracts.	Developers (IT)	Legality
43	[81]	Oracle trustworthiness and compliance in legal adjudication	A novel framework for secure and efficient oracle development.	Developers (IT)	Legality

No	Study	Concern	Interventions	Stakeholders	Area
44	[19]	Smart contracts lack clear legal regulation; applying current law is challenging	Clarification of SC in relation to the Civil Code	Developers (IT & non-IT) & Users	Legality
45	[66]	Updating SC and integrating blockchain in legal systems	System architecture with UI, application logic, and blockchain	Developers (IT)	Legality
46	[26]	The relationship between transaction accounting, immutable code trust, and leveraging distributed crowds and databases.	Examines the landscape of blockchain in terms of trust, governance, decentralisation	Developers (IT & non-IT) & Users	Ethical and Social
47	[73]	Fairness issues arising from the logical design.	FairCon, a framework for verifying fairness properties	Developers (IT & non-IT) & Users	Ethical & Social
48	[118]	Ethical and design challenges in using blockchain for social impact.	Human-based survey - Evaluating design choices.	Developers (IT & non-IT) & Users	Ethical & Social
49	[14]	Gaps in understanding the sociotechnical factors involved in developing SCs.	Analysis of sociotechnical dynamics in open-source repositories for SCs.	Developers (IT & non-IT) & Users	Ethical & Social
50	[9]	Can blockchain SCs transform the social contract to address issues like corruption?	Introducing the smart social contract by integrating blockchain with social contract philosophy.	Developers (IT)	Ethical & Social
51	[63]	Challenges users encounter when interacting with blockchain technology.	Human-based survey - Usability testing for dApp Application	Users	Usability/ Cost & Fees
52	[34]	lack of systematic understanding of user-centered cryptocurrency threats	Human-based survey - Insights for understanding user-centered threats	Users	Usability/ Cost & Fees/ Trust
53	[51]	Non-tech-savvy end-users cannot interpret the source code.	SMARTDOC: Automatically generates user notices for smart contract functions	Users	Usability
54	[85]	Users lack essential knowledge to avoid vulnerable and malicious contract code	Tx2TXT: Automatically generates security-centric textual descriptions	Users	Usability
55	[65]	Adversaries exploiting blockchain's pseudo-anonymity threaten accountability and attribution	A novel user-centric visualisation framework for transactions	Users	Usability
56	[131]	The difficulty users face in understanding and securely managing SC wallets	Human-based survey - Users interaction with SC wallets	Users	Usability
57	[75]	Accessibility for users with disabilities.	Human-based survey - Analysing accessibility	Users	Usability
58	[64]	Accessibility and understandability concerns, especially for vulnerable individuals	Visualisation platform for creating and verifying contract content.	Users	Usability

No	Study	Concern	Interventions	Stakeholders	Area
59	[80]	Developing public understanding of technologies	Human-based survey - Explaining blockchain from an HCI perspective.	Users	Usability
60	[104]	The responsibility placed on users to manage their own security in Web3-based DApps.	Human-based survey - Primary human security concerns and key mitigation strategies	Users	Usability
61	[33]	SCs are complex and not user-friendly for the average user.	SC generator using Ethereum's ERC standards with a configuration wizard	Users	Human-Readability
62	[126]	To assist non-programmers in comprehending the logic of smart contracts.	SmartBT to generate readable function descriptions from SC bytecode to improve user understanding.	Users	Human-Readability /Usability
63	[83]	Complex gas triangle should not be exposed to end-users	Empirical evidence to support the claim.	Users	Cost & Fees
64	[29]	The impact of fee prices on user activities on Ethereum.	Empirical study - Insights and analysis of fees	Users	Cost & Fees
65	[70]	Centralisation risk associated with smart contracts.	Insights on the implications of centralisation risk	Users	Governance/Trust
66	[62]	Centralisation risk associated with smart contracts.	A library to ensure responsible ownership and management of ERC20 tokens	Users	Governance
67	[128]	Centralisation risk associated with smart contracts.	Empirical study - Detecting centralised security risks	Users	Governance
68	[37]	Privileged parties at the application layer.	Ethpector tool for detecting privileged parties in binary smart contract code on Ethereum	Users	Governance/Trust
69	[15]	Human trust in these systems is an issue	Human-based survey - Understand trust in blockchain systems	Users	Trust
70	[7]	Examines prosumers' concerns about smart contracts	Human-based survey - Social and legal acceptance.	Users	Trust
71	[127]	The clash between decentralisation ideals & issues like hacking and counterfeit NFTs	An exploratory study analysed stakeholder concerns in the NFT market	Users	Trust
72	[55]	Upgradeable contracts can change functions unnoticed, causing user distrust.	Bytecode-based classification model to detect contracts using the proxy upgrade pattern.	Users	Trust
73	[53]	Investors and creators face trust and fairness issues due to NFT market manipulation.	Large-scale data analysis of NFT transactions and marketplaces	Users	Trust
74	[35]	Challenges faced by early users in the Cryptocurrency.	Human-based survey- Identifying challenges for the HCI community.	Users	Usability/ Cost & Fees/ Trust

No	Study	Concern	Interventions	Stakeholders	Area
75	[31]	The need for human-centric, sustainable and resilient smart contracts in Industry 5.0.	An overview and comparative analysis of blockchain technologies in the context of Industry 5.0.	Developers (IT) & Users	Usability/ Ethical & Social
76	[84]	The complexity of Layer 2 integration increases development challenges and limits user usability.	An analysis study examines fees, usability and solutions for Ethereum DApps.	Developers (IT) & Users	Language /Usability/ Cost
77	[20]	High transaction fees limit users' accessibility, usability, and the sustainability of decentralised social media.	Human-based survey to analyse transaction fees and user acceptance	Users	Usability/ Cost

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